COMPLETE RESEARCH EDITION

Beukers–Delaygue Supercongruences, Apéry Truncations, and the Level-54 Attractor

Prepared for Keio

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Abstract. We study the precision of truncated hypergeometric periods under Frobenius. For powers of the central binomial coefficients, finite reflection followed by an integral lift of the digit operator proves the fixed-precision and all-level statements of Beukers and Delaygue. An elliptic crystal gives a two-step congruence for Apéry truncations on smooth supersingular disks. At the level-54 attractor of AESZ4 we construct a tetrahedral projector, identify its two-dimensional image with the weight-four modular factor, and interpolate the factorial blocks to recover the truncated period modulo p^3. The resulting argument applies at every good prime, including primes for which the modular coefficient is divisible by p.

Principal results and their hypotheses

Reflection and all lengths Parts I–II, pp. 4, 24

For odd p, $d \geq 2$, $\epsilon = \pm 1$, and $F_p(\epsilon) \in \mathbb{Z}_p^\times$, the exponent is $2s$ for every length mp^s with $m, s \geq 1$ (Part II, Theorem 1.1).

Supersingular Apéry disks Part I, p. 17

For $p > 3$, $p^2 \mid N$, and the smoothness and supersingularity conditions of Theorem 6.1, the two-step defect is divisible by $p^{3v_p(N)-1}$.

The AESZ4 special value Parts III–IV, pp. 32, 42

Part III determines the cutoff residual and the generic residual monodromy. Part IV, Theorem 1.1, gives $S_p \equiv a_p \pmod{p^3}$ for $p \geq 5$ at $t = -1/8$. The printed quantifier $p \neq 3$ includes $p = 2$, where $S_2 = 161/162$ and $a_2 = 2$.

Section, equation and appendix numbers are local to each part. All four parts share a bibliography. The new arguments in this research manuscript await independent external review.

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PART I

Reflection Operators and Supersingular Apéry Supercongruences

Research manuscript prepared for Keio

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Abstract

For the truncations $F_L(\epsilon) = \sum_{n<L}((1/2)_n/n!)^d\epsilon^n$, we prove the full mathematical statement of Beukers–Delaygue's Conjecture 1.2 by a finite reflection calculation. Retaining this reflection in an integral analytic digit operator gives the stronger, all-length estimate

$$F_{mp^s}(\epsilon) - f_p(\epsilon)F_{mp^{s-1}}(\epsilon) \in p^{\min(2s,s+d-1)}\mathbb{Z}_p$$

for every odd prime p, $d \geq 2$, $\epsilon = \pm 1$ with $F_p(\epsilon)$ a unit, and $m, s \geq 1$. It proves the precision predicted by their Conjecture 1.5 throughout $s \leq d-1$, and Part II, Theorem 1.1, extends it to every higher level. Separately, for the Apéry polynomials $P_N(t)$, an integral modular trace and an explicit elliptic crystal yield

$$P_N(t_0) - p^2 P_{N/p^2}(t_0) \in p^{3v_p(N)-1}\mathbb{Z}_p.$$

This holds for every $p^2 \mid N$ on each smooth supersingular residue disk specified in Theorem 6.1, for $p > 3$. The latter theorem includes the two rational CM values $t_0 = \pm 1$ under their respective inert-prime conditions and also non-CM lifts. Complete proofs and exact finite verification programs are included.

Scope and research status. Conjecture 1.2 of [1] is proved in its full stated range. Conjecture 1.5 is proved here for $1 \leq s \leq d-1$; for $s \geq d$ the argument in this part gives $s + d - 1$. The lift in Part II gives $2s$ throughout. The Apéry theorem has the explicit smoothness and supersingularity hypotheses of Theorem 6.1. The arguments have received internal independent checks; no completed external peer review or claim of literature priority is asserted. No unpublished BV3 or Swisher manuscript is assumed.

Reading guide. Sections 1–4 prove the fixed-modulus reflection theorem. Section 5 supplies the stronger analytic operator and its all-length precision bound. Section 6 gives the independent Apéry trace, integral jet comparison and supersingular specialization. The appendices contain supplementary exact checks; none is used as a substitute for a proof.

1 Statements and relation to the published conjecture

Throughout, p is an odd prime and d is an integer with $d \geq 2$. Set

$$\alpha_k = \frac{(1/2)_k}{k!} = \frac{\binom{2k}{k}}{4^k}, \qquad F_L(z) = \sum_{k=0}^{L-1} \alpha_k^d z^k \quad (L \geq 1).$$

In particular, $F_1(z) = 1$. All coefficients belong to $\mathbb{Z}_p$, since their displayed denominators are powers of 4. Write

$$N = \frac{p-1}{2}, \qquad \chi = \chi_{p,d}(\epsilon) = (-1)^{dN}\epsilon^N, \qquad \epsilon \in \{1, -1\}. \tag{1}$$

Theorem 1.1 (Block congruence)**.** *If $\chi = 1$, then for every integer $m \geq 1$,*

$$F_{mp}(\epsilon) \equiv F_p(\epsilon)F_m(\epsilon) \pmod{p^2\mathbb{Z}_p}. \tag{2}$$

No unit assumption on $F_p(\epsilon)$ is required for this statement.

The sign condition holds for both values of ϵ if $p \equiv 1 \pmod 4$. If $p \equiv 3 \pmod 4$, it holds exactly when $\epsilon = (-1)^d$.

Theorem 1.2 (The Beukers–Delaygue unit-root congruence)**.** *Suppose $F_p(\epsilon) \in \mathbb{Z}_p^\times$. The Dwork limit*

$$f_p(\epsilon) = \lim_{s \longrightarrow \infty} \frac{F_{p^s}(\epsilon)}{F_{p^{s-1}}(\epsilon)} \tag{3}$$

exists, and

$$F_p(\epsilon) \equiv f_p(\epsilon) \pmod{p^2\mathbb{Z}_p}. \tag{4}$$

The existence of (3) is the published Dwork input recorded in the introduction of [1]; it is not a new assertion of this note. The congruence (4) is exactly their Conjecture 1.2. Their Theorem 1.1 proves it at $\epsilon_p = (-1)^{d(p-1)/2}$, and their Corollary 2.4 excludes the opposite sign from the unit case when $p \equiv 3 \pmod 4$. They identify $p \equiv 1 \pmod 4$, $\epsilon = -1$ as the remaining case. The reflection calculation below applies to that case as well.

2 Elementary congruences

The calculations in this section also give self-contained proofs of the elementary inputs appearing as Lemmas 2.1–2.5 of [1].

Lemma 2.1 (Babbage congruence)**.** *For integers $A \geq B \geq 0$,*

$$\binom{pA}{pB} \equiv \binom{A}{B} \pmod{p^2}.$$

Proof. The boundary cases are immediate. In the product for $\binom{pA}{pB}$, separate the factors indexed by multiples of p:

$$\binom{pA}{pB} = \binom{A}{B} \prod_{\substack{1 \leq j \leq p(A-B) \\ p \nmid j}} \left(1 + \frac{pB}{j}\right).$$

The second factor is congruent modulo p^2 to $1 + pB\sum_{1 \leq j \leq p(A-B),\, p \nmid j} j^{-1}$. Each complete block contributes $\sum_{a=1}^{p-1} a^{-1} = 0$ in $\mathbb{F}_p$: inversion permutes the nonzero residues, whose sum is zero for odd p. The product is therefore 1 modulo p^2. □

Put

$$h_t = \sum_{j=1}^{2t} \frac{(-1)^{j-1}}{j} \quad (0 \le t \le N), \qquad \gamma = \frac{4^{p-1}-1}{p} \in \mathbb{Z}. \tag{5}$$

The empty sum defining h_0 is zero.

Lemma 2.2 (Alternating harmonic sum)**.** *One has*

$$\sum_{j=1}^{p-1} \frac{(-1)^{j-1}}{j} \equiv \gamma \pmod{p}.$$

Proof. The identity

$$\frac{2^p - 2}{p} = \sum_{j=1}^{p-1} \frac{1}{j} \binom{p-1}{j-1}$$

and $\binom{p-1}{j-1} \equiv (-1)^{j-1} \pmod{p}$ show that the left side of the desired congruence equals $(2^p-2)/p$ modulo p. If $q = (2^{p-1}-1)/p$, then $\gamma = q(2^{p-1}+1) \equiv 2q = (2^p-2)/p \pmod{p}$. □

Lemma 2.3 (Two reflections)**.** *For $0 \le t \le N$,*

$$\alpha_{N-t} \equiv (-1)^N \alpha_t \pmod{p}, \tag{6}$$

$$h_{N-t} \equiv \gamma - h_t \pmod{p}. \tag{7}$$

Consequently, for $w_t = \alpha_t^d \epsilon^t$,

$$w_{N-t} \equiv \chi w_t \pmod{p}. \tag{8}$$

Proof. Since $1/2 \equiv -N \pmod{p}$ and $t!$ is a unit,

$$\alpha_t \equiv \frac{(-N)_t}{t!} = (-1)^t \binom{N}{t} \pmod{p}.$$

Binomial symmetry proves (6). For (7), subtract from the complete alternating sum the final $2t$ terms. Under $j = p-k$, those terms satisfy

$$\frac{(-1)^{j-1}}{j} = \frac{(-1)^{p-k-1}}{p-k} \equiv \frac{(-1)^{k-1}}{k} \pmod{p}.$$

Their sum is therefore h_t modulo p, and Lemma 2.2 supplies the complete sum. Finally $\epsilon^{N-t} = \epsilon^N \epsilon^t$, giving (8). □

Lemma 2.4 (Uniform digit expansion)**.** *Let $a \ge 0$ and $0 \le t < p$ be integers. If $t > N$, then $\alpha_{pa+t} \in p\mathbb{Z}_p$. If $t \le N$, then*

$$\alpha_{pa+t} \equiv \alpha_a \alpha_t (1 + pa(2h_t - \gamma)) \pmod{p^2 \mathbb{Z}_p}. \tag{9}$$

There is no upper bound on a in this assertion.

Proof. The exact identity

$$\binom{2pa+2t}{pa+t} = \binom{2pa}{pa} \frac{\prod_{j=1}^{2t}(2pa+j)}{\prod_{j=1}^{t}(pa+j)^2} \tag{10}$$

has unit denominator in its second factor. If $t > N$, its numerator contains the factor $2pa + p$, proving the divisibility assertion. If $t \le N$, all displayed j are prime to p, and the second factor is

$$\binom{2t}{t}\left(1 + 2pa\left(\sum_{j=1}^{2t}\frac{1}{j} - \sum_{j=1}^{t}\frac{1}{j}\right)\right) \equiv \binom{2t}{t}(1 + 2pah_t) \pmod{p^2}.$$

Apply Lemma 2.1 to the first binomial coefficient in (10). After division by 4^{pa+t}, the remaining normalization factor is

$$4^{a(1-p)} = (1 + p\gamma)^{-a} \equiv 1 - pa\gamma \pmod{p^2}.$$

Multiplying these expressions proves (9). □

3 Cancellation and the block theorem

Lemma 3.1 (Cancellation of the block correction)**.** *If $\chi = 1$, then*

$$\sum_{t=0}^{N}(2h_t - \gamma)\alpha_t^d\epsilon^t \equiv 0 \pmod{p}. \tag{11}$$

If $F_p(\epsilon)$ is a p-adic unit, then necessarily $\chi = 1$.

Proof. Let $S = \sum_{t=0}^{N} h_t w_t$ and $A = \sum_{t=0}^{N} w_t$. When $\chi = 1$, reflection and Lemma 2.3 give

$$S = \sum_{t=0}^{N} h_{N-t}w_{N-t} \equiv \sum_{t=0}^{N}(\gamma - h_t)w_t = \gamma A - S \pmod{p}.$$

Thus $2S - \gamma A \equiv 0$, which is (11). For the second assertion, Lemma 2.4 with $a = 0$ implies

$$F_p(\epsilon) \equiv A \pmod{p^2}, \tag{12}$$

because $d \ge 2$. If $\chi = -1$, reflection gives $A \equiv -A \pmod{p}$. Since p is odd, this would imply $F_p(\epsilon) \equiv 0 \pmod{p}$. □

Proof of Theorem 1.1. Write every summation index uniquely as $pa + t$, with $0 \le a < m$ and $0 \le t < p$. By Lemma 2.4, the terms with $t > N$ vanish modulo p^2 after taking the dth power. The remaining terms satisfy

$$\alpha_{pa+t}^d\epsilon^{pa+t} \equiv \alpha_a^d\epsilon^a\alpha_t^d\epsilon^t(1 + pda(2h_t - \gamma)) \pmod{p^2},$$

where $\epsilon^{pa} = \epsilon^a$ because p is odd. Summing yields

$$\begin{aligned} F_{mp}(\epsilon) \equiv F_m(\epsilon)&\sum_{t=0}^{N}\alpha_t^d\epsilon^t \\ &+ pd\left(\sum_{a=0}^{m-1} a\alpha_a^d\epsilon^a\right)\left(\sum_{t=0}^{N}(2h_t - \gamma)\alpha_t^d\epsilon^t\right) \pmod{p^2}. \end{aligned}$$

The second line vanishes by Lemma 3.1, and the first line is $F_m(\epsilon)F_p(\epsilon)$ by (12). □

Corollary 3.2. *If $\chi = 1$, then for every $s \ge 1$,*

$$F_{p^s}(\epsilon) \equiv F_p(\epsilon)F_{p^{s-1}}(\epsilon) \equiv F_p(\epsilon)^s \pmod{p^2}.$$

Proof. Use $m = p^{s-1}$ in Theorem 1.1 and then induct on s. □

4 Passage to the unit-root limit

Proof of Theorem 1.2. By Lemma 3.1, the unit assumption implies $\chi = 1$. Corollary 3.2 shows that every $F_{p^s}(\epsilon)$ is a unit and that, for $s \geq 1$,

$$R_s := \frac{F_{p^s}(\epsilon)}{F_{p^{s-1}}(\epsilon)} \equiv F_p(\epsilon) \pmod{p^2}. \tag{13}$$

The Dwork congruences as stated in [1, Introduction, equation (1)] give $R_{s+1} \equiv R_s \pmod{p^s}$; here $\epsilon^p = \epsilon$. Hence the sequence (R_s) converges in $\mathbb{Z}_p$ to the value denoted by $f_p(\epsilon)$ in (3). Since $p^2\mathbb{Z}_p$ is closed, taking the limit in (13) proves (4). □

Remark 4.1 (What changes in the published proof). The block expansion and the paired harmonic-sum calculation are the method of [1, Section 2]. The additional observation is to retain the joint factor $\chi = (-1)^{dN}\epsilon^N$ in the reflection, instead of specializing in advance to $\epsilon = \epsilon_p$. For $p \equiv 1 \pmod 4$, the integer N is even, so this factor is 1 also at $\epsilon = -1$. Thus the same cancellation completes the missing sign without any new modular, crystalline, or unpublished input.

Remark 4.2 (Fixed precision versus growing precision). The conclusion for every level in Corollary 3.2 has the fixed modulus p^2. It does not supply $F_{p^s}(\epsilon) - f_p(\epsilon)F_{p^{s-1}}(\epsilon) \in p^{2s}\mathbb{Z}_p$. The latter assertion is Conjecture 1.5 of [1]. Section 5 proves it for $s \leq d-1$ and obtains the weaker exponent $s+d-1$ for $s \geq d$. We do not identify $f_p(\epsilon)$ with a particular modular-form coefficient or an explicitly specified motivic Euler factor.

For a concrete instance of the added sign, take $p = 5$, $d = 2$, and $\epsilon = -1$. Direct rational arithmetic gives

$$F_5(-1) = 1 - \frac{1}{4} + \frac{9}{64} - \frac{25}{256} + \frac{1225}{16384} = \frac{14217}{16384} \equiv 13 \pmod{25}.$$

This is a unit, so $f_5(-1) \equiv 13 \pmod{25}$ by Theorem 1.2. The block theorem also gives $F_{25}(-1) \equiv 13^2 \equiv 19 \pmod{25}$.

5 A precision bound for every level and every length

The reflection can also be retained in an integral analytic operator. Keeping track of its even and odd parts yields a stronger result than the fixed-modulus congruence. Put

$$\nu_d(s) = \min(2s, s+d-1) = s + \min(s, d-1).$$

Theorem 5.1 (All-length precision bound). *Let p be any odd prime, $d \geq 2$, and $\epsilon \in \{1, -1\}$. Suppose $F_p(\epsilon) \in \mathbb{Z}_p^\times$, and let $f_p(\epsilon)$ be the limit in (3). For all integers $m, s \geq 1$,*

$$F_{mp^s}(\epsilon) - f_p(\epsilon)F_{mp^{s-1}}(\epsilon) \in p^{\nu_d(s)}\mathbb{Z}_p. \tag{14}$$

No unit assumption on $F_m(\epsilon)$ is needed. Equivalently, if $K \geq 1$ and $k = v_p(K) \geq 1$, then

$$F_K(\epsilon) - f_p(\epsilon)F_{K/p}(\epsilon) \in p^{k+\min(k,d-1)}\mathbb{Z}_p. \tag{15}$$

Corollary 5.2 (A range of the growing-precision conjecture). *Under these assumptions, for every $1 \leq s \leq d-1$,*

$$F_{p^s}(\epsilon) \equiv f_p(\epsilon)F_{p^{s-1}}(\epsilon) \pmod{p^{2s}}.$$

Thus the sth assertion of Beukers–Delaygue's Conjecture 1.5 is proved for all $d \geq s+1$. At levels $s \geq d$, Theorem 5.1 instead supplies the modulus p^{s+d-1}.

The proof below is elementary. In particular, the exact reflection of the interpolating functions is proved from finite products; no property of the p-adic gamma function is required.

5.1 Integral analytic interpolation of digit blocks

Let $\mathcal{A} = \mathbb{Z}_p\langle a\rangle$ be the algebra of restricted power series $\sum_{j\ge 0} b_j a^j$, where $b_j \in \mathbb{Z}_p$ and $b_j \to 0$. We use its Gauss norm. Divisibility in $\mathcal{A}$ always means divisibility of every power-series coefficient. Integral affine substitution, evaluation at a point of $\mathbb{Z}_p$, and finite integral sums are bounded operations on this algebra.

Lemma 5.3 (Analytic digit functions)**.** *There are functions $g_t \in \mathcal{A}$, $0 \le t < p$, such that for every nonnegative integer a,*

$$g_t(a) = \frac{\alpha_{pa+t}}{\alpha_a}. \tag{16}$$

For $0 \le t \le N$, there is $b_t \in \mathbb{Z}_p$ with

$$g_t(a) \equiv \alpha_t(1 + pb_t a) \pmod{p^2\mathcal{A}}. \tag{17}$$

For $N < t < p$, one has $g_t \in p\mathcal{A}$. All these statements hold also for $p = 3$.

Proof. First interpolate $g_0(a) = \alpha_{pa}/\alpha_a$. For nonnegative integer a, cancellation of the numerator and denominator factors divisible by p gives

$$\frac{g_0(a+1)}{g_0(a)} = R(a), \qquad R(a) = C\,\frac{\prod\limits_{\substack{0\le t<p\\ t\ne N}} (1 + pa/(t+1/2))}{\prod\limits_{i=1}^{p-1}(1+pa/i)}, \qquad C = \frac{\prod\limits_{\substack{0\le t<p\\ t\ne N}} (t+1/2)}{\prod\limits_{i=1}^{p-1} i}.$$

Both constant-factor lists reduce to $\mathbb{F}_p^\times$, so $C \in 1 + p\mathbb{Z}_p$. For $j \ge 1$ put

$$s_j = \sum_{\substack{0\le t<p\\ t\ne N}} (t+1/2)^{-j} - \sum_{i=1}^{p-1} i^{-j} \in p\mathbb{Z}_p.$$

The logarithm, defined by its convergent power series on $1 + p\mathcal{A}$, is

$$\log R(a) = \log C + \sum_{j\ge 1} \frac{(-1)^{j-1} p^j s_j}{j} a^j. \tag{18}$$

The polynomial $P_j(a) = \sum_{u=0}^{a-1} u^j$, initially defined on nonnegative integers, has the expression

$$P_j(a) = \sum_{v=0}^{j} \left\{ {j \atop v} \right\} \frac{a(a-1)\cdots(a-v)}{v+1}.$$

Here the braces denote Stirling numbers of the second kind. Consequently every power-basis coefficient of P_j has valuation at least $-\lfloor \log_p(j+1) \rfloor$. Define

$$L(a) = a \log C + \sum_{j\ge 1} \frac{(-1)^{j-1} p^j s_j}{j} P_j(a). \tag{19}$$

The coefficients contributed by its jth summand have valuation at least

$$j + 1 - v_p(j) - \lfloor \log_p(j+1) \rfloor \ge 2, \tag{20}$$

and this lower bound tends to infinity. For completeness, the inequality is immediate for $j = 1, 2, 3$. For $j \geq 4$ use $p \geq 3$ and $3^{j-1} \geq j(j+1)$; the latter holds at $j = 4$ and follows thereafter by induction. Thus (19) converges in $\mathcal{A}$, $L \in p\mathcal{A}$, and $L - a \log C \in p^2\mathcal{A}$.

Since p is odd, the exponential converges on $p\mathcal{A}$. Set $g_0 = \exp L$. The identities $P_j(a+1) - P_j(a) = a^j$ give $g_0(a+1) = R(a)g_0(a)$ and $g_0(0) = 1$. These identify its nonnegative integer values with α_{pa}/α_a. Moreover, $g_0(a) \equiv 1 + a \log C \pmod{p^2\mathcal{A}}$. For the remaining digits set

$$g_t(a) = g_0(a) 4^{-t} \frac{\prod_{j=1}^{2t}(2pa + j)}{\prod_{j=1}^{t}(pa + j)^2}. \tag{21}$$

All denominator constants are units, so this expression lies in $\mathcal{A}$. The exact binomial product (10) proves (16). If $t \leq N$, expansion of the unit factors proves (17), with constant term α_t. If $t > N$, the numerator has the factor $p(2a + 1)$, proving $g_t \in p\mathcal{A}$. □

5.2 An exact reflection from finite products

Lemma 5.4 (Exact analytic reflection)**.** *Let $r(a) = -a - 1/2$. For every $0 \leq t \leq N$,*

$$g_{N-t}(r(a)) = (-1)^N g_t(a) \qquad \text{in } \mathcal{A}. \tag{22}$$

Proof. Put $N_j = (p^j - 1)/2$. For each fixed nonnegative integer n and all sufficiently large j, a finite product gives

$$\frac{\alpha_{N_j - n}}{\alpha_{N_j}} = \prod_{k=0}^{n-1} \frac{N_j - k}{N_j - k - 1/2} \longrightarrow \prod_{k=0}^{n-1} \frac{k + 1/2}{k + 1} = \alpha_n.$$

The convergence follows from $N_j \to -1/2$ in $\mathbb{Z}_p$; every limiting denominator is nonzero in $\mathbb{Q}_p$. Also, since $N_j = pN_{j-1} + N$,

$$\frac{\alpha_{N_j}}{\alpha_{N_{j-1}}} = g_N(N_{j-1}) \longrightarrow C_* := g_N(-1/2).$$

Fix a nonnegative integer a and $0 \leq t \leq N$, and set $n = pa + t$, $a_j = N_{j-1} - a$. For large j, $pa_j + (N - t) = N_j - n$. Applying the two finite-product limits above to the numerator and denominator of $g_{N-t}(a_j) = \alpha_{N_j - n}/\alpha_{N_{j-1} - a}$ yields

$$g_{N-t}(r(a)) = C_* \frac{\alpha_n}{\alpha_a} = C_* g_t(a).$$

Continuity extends this equality from nonnegative integers to $\mathbb{Z}_p$. It is also an identity of restricted power series: a nonzero convergent power series cannot vanish at all p^j, which approach zero. Applying the identity twice gives $C_*^2 = 1$, using $g_0(0) = 1$. By (17) and (6), $C_* \equiv \alpha_N \equiv (-1)^N \pmod{p}$. The two roots $1, -1$ are distinct modulo odd p, so $C_* = (-1)^N$. □

5.3 Digit operators and their two reflection components

Define bounded integral operators on $\mathcal{A}$ by

$$(Sf)(a) = \sum_{t=0}^{p-1} \epsilon^t g_t(a)^d f(pa + t), \tag{23}$$

$$(T_0 f)(a) = \sum_{t=0}^{N} \epsilon^t g_t(a)^d f(pa + t). \tag{24}$$

By Lemma 5.3,

$$S = T_0 + p^d K \tag{25}$$

for a bounded integral operator K. Let $\mathcal{R}f = f \circ r$. Throughout the remainder of the proof, the ordinary hypothesis and Lemma 3.1 give $\chi = 1$. Equation (22), together with $r(pa + t) = pr(a) + (N - t)$, then gives

$$T_0\mathcal{R} = \mathcal{R}T_0. \tag{26}$$

Lemma 5.5 (Operator divisibilities)**.** *Set $c = -1/4$, the fixed point of r, and $\ell(f) = f(c)$. Decompose*

$$\mathcal{A} = \mathbb{Z}_p \cdot 1 \ \oplus \ E \ \oplus \ O,$$

where $E = \{f : \mathcal{R}f = f, \ \ell(f) = 0\}$ and $O = \{f : \mathcal{R}f = -f\}$. In this decomposition the block valuations of S are at least

$$\begin{pmatrix} 0 & 0 & d \\ 2 & 2 & d \\ d & d & 1 \end{pmatrix}. \tag{27}$$

An entry b means that the corresponding component map has image divisible by p^b. The scalar entry $\ell(S1)$ is a unit.

Proof. The projections $(1 \pm \mathcal{R})/2$ are bounded and integral. Every odd function vanishes at c, so this is indeed a bounded direct sum decomposition. For arbitrary $f \in \mathcal{A}$, the congruence

$$f(pa + t) \equiv f(t) + paf'(t) \pmod{p^2\mathcal{A}}$$

and (17) show that $T_0 f$ is affine modulo p^2. If f is even, (26) makes $T_0 f$ even. An affine polynomial $A + Ba$ fixed by $a \mapsto -a - 1/2$ modulo p^2 has $B \in p^2\mathbb{Z}_p$. Thus its nonconstant even component is divisible by p^2. If f is odd, $T_0 f$ is odd and is constant modulo p; that constant must be zero, so $T_0 f \in pO$. These observations give (27) for T_0, with zero off-diagonal reflection components. Equation (25) gives the displayed matrix for S, since $d \geq 2$. Finally $S1$ is constant modulo p with value $\sum_{t=0}^{N} \alpha_t^d \epsilon^t \equiv F_p(\epsilon)$, proving the unit assertion. □

5.4 Normalized iteration and its precision

Lemma 5.6 (Iteration estimate)**.** *There is a unit $\lambda \in \mathbb{Z}_p^\times$ such that, for all $s \geq 1$,*

$$S^s 1 - \lambda S^{s-1} 1 \in p^{\nu_d(s)}\mathcal{A}. \tag{28}$$

Proof. On the set $\mathcal{Q} = 1 + p^2 E + p^d O$ define

$$\mathcal{N}(h) = \frac{Sh}{\ell(Sh)}.$$

The denominator is a unit by (27), and the same matrix shows that $\mathcal{N}$ maps $\mathcal{Q}$ into itself. Let $h_0 = 1$, $h_s = \mathcal{N}(h_{s-1})$, and $Z_s = \ell(S^s 1)$. Induction gives $Z_s \in \mathbb{Z}_p^\times$ and $h_s = S^s 1/Z_s$.

For $g, h \in \mathcal{Q}$ the exact identity

$$\mathcal{N}(g) - \mathcal{N}(h) = \frac{S(g - h) - \ell(S(g - h))\mathcal{N}(h)}{\ell(Sg)} \tag{29}$$

shows the following. If the even and odd parts of $g - h$ belong to $p^u E$ and $p^v O$, respectively, then the parts of the left side belong to

$$p^{\min(u+2, v+d)} E, \qquad p^{\min(u+d, v+1)} O.$$

Indeed, (27) gives these bounds for $S(g-h)$; the subtracted scalar has valuation at least $\min(u, v+d)$ and multiplies even and odd components divisible by p^2 and p^d.

Put $\Delta_s = h_s - h_{s-1}$. Starting with $\Delta_1 \in p^2E + p^dO$, induction in (29) gives

$$\Delta_s \in p^{u_s}E + p^{v_s}O, \qquad u_s = \min(2s, 2d+s-2), \quad v_s = d+s-1. \tag{30}$$

For example, the inductive lower bounds are $\min(u_{s-1}+2, v_{s-1}+d) = u_s$ and $\min(u_{s-1}+d, v_{s-1}+1) = v_s$ for $s \geq 2$. Thus h_s converges in $\mathcal{A}$ to $h_\infty \in \mathcal{Q}$.

Set

$$z_s = \frac{Z_s}{Z_{s-1}} = \ell(Sh_{s-1}), \qquad \lambda = \ell(Sh_\infty) \in \mathbb{Z}_p^\times.$$

The scalar row of (27) gains an additional d powers of p on odd inputs. Since $h_\infty - h_{s-1} = \sum_{j=s}^{\infty} \Delta_j$, it follows that

$$\lambda - z_s \in p^{u_s}\mathbb{Z}_p. \tag{31}$$

Here u_j is nondecreasing and $\min(u_j, v_j + d) = u_j$. Consequently,

$$S^s 1 - \lambda S^{s-1}1 = Z_{s-1}\big(z_s(h_s - h_{s-1}) + (z_s - \lambda)h_{s-1}\big).$$

Equations (30)–(31) give valuation at least $\min(u_s, v_s) = \min(2s, s+d-1) = \nu_d(s)$ throughout this function. This proves (28). □

5.5 Evaluation at the original truncations

Proof of Theorem 5.1. For $m \geq 1$ define the bounded integral functional

$$\mathcal{L}_m(f) = \sum_{a=0}^{m-1} \alpha_a^d \epsilon^a f(a).$$

Exact digit expansion (16) gives

$$\mathcal{L}_m(S^s 1) = F_{mp^s}(\epsilon) \qquad (s \geq 0). \tag{32}$$

To verify this, an iterated digit sequence represents a unique b with $0 \leq b < p^s$. The product of its g_t factors telescopes to $\alpha_{p^s a+b}/\alpha_a$. Its sign, after multiplication by ϵ^a, is $\epsilon^{p^s a+b}$, because p is odd. Summing over a and b gives (32). This identity uses rational equalities and does not require α_a or $F_m(\epsilon)$ to be a unit.

For $m = 1$, $F_{p^s}(\epsilon) = (S^s 1)(0) = Z_s h_s(0)$. Every $h_s(0)$ is a unit since $h_s \in 1 + p^2E + p^dO$. Therefore (28), evaluated at zero, implies

$$\frac{F_{p^s}(\epsilon)}{F_{p^{s-1}}(\epsilon)} \equiv \lambda \pmod{p^{\nu_d(s)}}.$$

Since $\nu_d(s) \to \infty$, the limit equals λ. It is precisely the limit $f_p(\epsilon)$ defined in (3). In particular, this operator argument also proves the convergence needed here. Applying $\mathcal{L}_m$ to (28) now proves (14), with the same $\lambda = f_p(\epsilon)$ for every m. Finally, write $K = mp^k$ with $p \nmid m$ to obtain (15). □

Remark 5.7 (The original evaluation point). Normalization took place at $c = -1/4$ only to separate reflection-even and reflection-odd functions. The sums themselves are evaluated at $a = 0$, as (32) shows. In particular,

$$\frac{F_{p^s}(\epsilon)}{F_{p^{s-1}}(\epsilon)} = z_s \frac{h_s(0)}{h_{s-1}(0)}.$$

The second quotient differs from 1 by an element of $p^{\nu_d(s)}\mathbb{Z}_p$ by (30), while $z_s - \lambda \in p^{u_s}\mathbb{Z}_p$ by (31). Thus centering the normalization does not assert an unjustified identification of evaluations at c and at zero.

Remark 5.8 (Scope of the higher precision). Under the unit hypothesis, the reflection estimate gives the precision of Conjecture 1.2 and the exponent $2s$ of Conjecture 1.5 for $1 \le s \le d-1$. For $s \ge d$ it gives $s + d - 1$. The lifted reflection of Part II, Theorem 1.1, gives $2s$ for these levels as well.

6 Apéry truncations on supersingular residue disks

In this section the notation F refers to the Apéry period. Put

$$A_n = \sum_{k=0}^{n} \binom{n}{k}^2 \binom{n+k}{k}^2, \qquad F(t) = \sum_{n\ge 0} A_n t^n, \qquad P_K(t) = \sum_{0\le n<K} A_n t^n.$$

We prove a congruence for every integral point in each smooth supersingular residue disk of the Apéry family. The proof first establishes an all-cutoff formal congruence, then identifies an explicit integral rank-two elliptic crystal, and finally specializes a transported rank-three jet identity. Its crystalline inputs are standard; the trace induction, differential residual, integral gauge and precision comparison are proved here. The statement is an all-level extension of first-level modular supercongruences, not a claim that the older Apéry coefficient congruences were previously open.

6.1 The exact full-cutoff residual

Set

$$L = \theta^3 - t(2\theta+1)(17\theta^2+17\theta+5) + t^2(\theta+1)^3, \qquad \theta = t\frac{d}{dt}.$$

The Apéry recurrence gives the exact identity

$$LP_K = K^3 t^K(-A_K + tA_{K-1}). \tag{33}$$

Indeed the coefficients below K vanish. At degree K the recurrence would have supplied K^3A_K, and at degree $K+1$ the remaining term is K^3A_{K-1}. There are no later terms. This proves (33) over $\mathbb{Z}[t]$, not merely modulo a prime power.

6.2 An independent integral trace proof

The published modular parametrization in Beukers, arXiv:2403.03301v3, Appendix C, Section 9.5 [7], is

$$t = \left(\frac{\eta_1\eta_6}{\eta_2\eta_3}\right)^{12}, \qquad \Theta = \frac{(\eta_2\eta_3)^7}{(\eta_1\eta_6)^5}, \qquad F(t(q)) = \Theta(q).$$

Here $\eta_j = \eta(j\tau)$, $q = e^{2\pi i\tau}$, $t \in q + q^2\mathbb{Z}[[q]]$, $\Theta \in 1 + q\mathbb{Z}[[q]]$, and the group is $G = \Gamma_0(6) + \langle W_6\rangle$. On $X_0(6)$, the cusp orders, in denominator order $1, 2, 3, 6$, are

c	1	2	3	6
$\operatorname{ord}_c t$	1	−1	−1	1
$\operatorname{ord}_c \Theta$	0	1	1	0
width	6	3	2	1

The eta transformation formula gives $t \circ W_6 = t$ and $\Theta|_2 W_6 = -\Theta$. The two pole cusps are exchanged by W_6. Since $t : X_0(6) \to \mathbb{P}^1$ has degree two, it induces a degree-one function on $X(G)$.

For $p > 3$, the integral determinant-p matrices with lower left entry divisible by six form the usual Hecke set. Every such matrix preserves the denominator class of a cusp modulo six. To see this prime by prime, reduce the matrix modulo 2 or 3: it is invertible and upper triangular, so it preserves whether the denominator is divisible by that prime. Cancelling a common factor can only remove a factor of p, which changes neither condition. Thus each of the two pole cusp classes is preserved. The Fricke normalizer preserves the Hecke set. Consequently the usual trace construction applies, with the same cusp width at the source and target of each branch.

For a polynomial $P \in \mathbb{Z}[T]$, put

$$Q_P(q) = \frac{\Theta(q)P(t(q^p))}{\Theta(q^p)}, \qquad G_P(q) = \frac{\Theta(q^p)P(t(q))}{\Theta(q)}.$$

With determinant-normalized weight-two slash operators, the sum of $p\Theta(\tau)P(t(\beta\tau))/(\Theta|_2\beta)(\tau)$ over the $p+1$ Hecke representatives equals

$$\mathrm{Tr}_p(P) = Q_P + p^3 U_p G_P. \tag{34}$$

The finite p representatives each contribute a factor p^2, and root-of-unity summation contributes the remaining p.

The trace is G-invariant. It has no poles away from the pole orbit of t, since Θ has no zeros there and the Hecke correspondence preserves that orbit in both directions. At a pole cusp, one branch multiplies local orders by p and the other p branches divide them by p. The respective orders of Θ, t are $1, -1$. It follows that

$$\mathrm{Tr}_p(P) = \Pi_P(t), \qquad \Pi_P \in \mathbb{Z}[T], \qquad \deg \Pi_P \le p \deg P + p - 1. \tag{35}$$

The coefficients are integral because (34) has an integral q-expansion and $t = q + O(q^2)$ has an integral inverse series.

Let $\sigma_p(t) = t(q(t)^p)$ and $\rho_p = F(t)/F(\sigma_p(t))$. Then, for all $m, s \ge 1$,

$$\rho_p(t) P_{mp^{s-1}}(\sigma_p(t)) - P_{mp^s}(t) \in p^{3s}\mathbb{Z}_p[[t]]. \tag{36}$$

Here is the complete induction. Apply (34) to $P = P_{mp^{s-1}}$. Its trace polynomial has degree $< mp^s$. By the preceding-level assertion,

$$G_P \equiv P_{mp^{s-2}}(t(q^p)) \pmod{p^{3(s-1)}}.$$

Thus $p^3 U_p G_P \equiv p^3 P_{mp^{s-2}}(t(q)) \pmod{p^{3s}}$. For $s = 1$ interpret the preceding polynomial as zero; the statement modulo p^3 is then automatic. Hence Q_P is congruent modulo p^{3s} to a polynomial of degree $< mp^s$. But $Q_P - F(t)$ has order at least mp^s in t. Comparing the initial mp^s coefficients identifies that polynomial with P_{mp^s} and proves (36).

6.3 An explicit smooth elliptic cover and integral gauge

Set

$$D = (1+8h)(1+9h), \qquad W = 1 - 72h^2, \qquad t = \frac{h}{D}.$$

Consider the elliptic curve

$$E_h : \quad y^2 = x^3 + \frac{1 + 12h + 24h^2}{4} x^2 + h^3(1+9h)x. \tag{37}$$

Its discriminant and c_4 are

$$\Delta = h^6(1+8h)^3(1+9h)^2, \qquad c_4 = (1+6h)(1+18h+84h^2+24h^3).$$

Thus it is smooth over $\mathbb{Z}_p[h, 1/(hD)]$ for every $p > 3$. With $\omega = dx/(2y)$ and $\eta = x\omega$, elementary differential reduction gives

$$\nabla_h \begin{pmatrix} \omega \\ \eta \end{pmatrix} = \begin{pmatrix} -6/(1+8h) & 6/(hD) \\ -6h^2/(1+8h) & 6/(1+8h) \end{pmatrix} \begin{pmatrix} \omega \\ \eta \end{pmatrix}. \tag{38}$$

For example, substitute $f = x^3 + ax^2 + bx$ and reduce $-f_h dx/(4y^3)$ modulo exact differentials $d(R/y)$. This is an identity of relative de Rham cohomology, with denominators only those displayed. In particular $(\omega, \nabla_h\omega)$ is an integral basis: its determinant relative to (ω, η) is $6/(hD)$, a unit.

The normalized vanishing-cycle period ℓ of ω at $h = 0$ has constant term one; the tangent lines at the node are $y = \pm x/2$. Equation (38) gives

$$\ell'' + \frac{1+34h+216h^2}{hD}\ell' + \frac{6(1+12h)}{hD}\ell = 0.$$

Hence $\ell = 1-6h+42h^2-312h^3+\cdots$, the twisted Zagier F period. The published eta parametrization is

$$h(q) = \frac{\eta_6^5\eta_2}{\eta_1^5\eta_3}, \qquad \ell(h(q)) = \frac{\eta_1^6\eta_6}{\eta_2^3\eta_3^2}.$$

Logarithmic eta differentiation gives the useful exact identity

$$q\frac{d}{dq}\log h = \frac{-5E_2(\tau) + 2E_2(2\tau) - 3E_2(3\tau) + 30E_2(6\tau)}{24} = \Theta = D\ell^2.$$

The last equality can also be checked in the weight-two modular form space for $\Gamma_0(6)$ by its coefficients through q^2. The elliptic Wronskian and the multiplicative fiber of order six give $d\log q_E/dh = 6/(hD\ell^2)$, while $\Delta \sim h^6$, $c_4 \sim 1$ fix the leading constant. Consequently the Tate parameter of (37) is exactly $q_E = q^6$. Canonical quotient of the elliptic curve sends q_E to q_E^p and therefore sends $h(q)$ to $h(q^p)$. This identifies the elliptic canonical lift with the modular lift used in the scalar trace argument. The eta identity, or a direct comparison of differential equations and initial terms, gives

$$F(t(h)) = D(h)\ell(h)^2. \tag{39}$$

No square root needs to be adjoined to construct the integral crystal. Use instead the quadratic twist

$$E_h^D: \quad Y^2 = D(h)\left(X^3 + \frac{1+12h+24h^2}{4}X^2 + h^3(1+9h)X\right), \qquad \omega_* = \frac{D(h)dX}{2Y}.$$

Its period is $a(h) = \sqrt{D}\,\ell(h)$, with the formal square root equal to one at $h = 0$, and $a^2 = F(t(h))$. As D is a unit, $(\omega_*, \nabla_h\omega_*)$ remains an integral basis. Explicitly, the period equation simplifies to

$$a'' + \mathcal{P}a' + \mathcal{Q}a = 0, \qquad \mathcal{P} = \frac{1}{h}, \quad \mathcal{Q} = -\frac{720h^2+169h+10}{4hD^2}. \tag{40}$$

Fix $m \geq 1$ with $(m, p) = 1$, put $K_s = mp^s$, and set $f_s(h) = P_{K_s}(t(h))$. Let $\mathcal{L}$ be the monic symmetric square of (40). Since

$$1 - 34t + t^2 = \frac{W^2}{D^2}, \qquad t' = \frac{W}{D^2},$$

the exact residual (33) becomes

$$\mathcal{L}f_s = K_s^3 \frac{W}{h^3 D}\, t^{K_s}(-A_{K_s} + tA_{K_s-1}). \tag{41}$$

This is integral on the whole smooth h-chart. In particular, no inverse of W is needed, despite the ramification of $h \mapsto t$.

6.4 The integral Frobenius–Taylor comparison

Write $A = \mathbb{Z}_p[\widehat{h, 1/(hD)}]$. Let ϕ be any integral Frobenius lift of this ring. In the integral basis $(\omega_*, \nabla_h\omega_*)$, use the row convention

$$F_\phi(\phi^*\omega_i) = \sum_j (M_\phi)_{ij}\omega_j, \qquad V_\phi = pM_\phi^{-1}.$$

Crystalline Frobenius and Verschiebung for a smooth elliptic family imply that both M_ϕ, V_ϕ are integral; their determinant is p times a unit. More explicitly, the cup product of $(\omega_*, \nabla_h\omega_*)$ is $6/h$ up to the fixed sign convention, so compatibility of Frobenius with cup product gives

$$\det M_\phi = p\frac{h}{\phi(h)}.$$

Thus the integrality of V_ϕ also follows directly from the adjugate of M_ϕ. The integral comparison and Frobenius are those for the Dieudonné crystal of the elliptic scheme's p-divisible group: Berthelot–Messing, *Théorie de Dieudonné cristalline I* [2], Section 2, Theorem 2 and the constructions on printed pp. 24–25; Section 3, equation (3.1), identifies this crystal with relative degree-one cohomology and its Hodge filtration. The square-period jet matrix is

$$S = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ -2\mathcal{Q} & -2\mathcal{P} & 2 \end{pmatrix}, \qquad B_\phi = S\,\mathrm{Sym}^2(V_\phi)(S^\phi)^{-1}.$$

Its determinant-changing matrices are integral units, since $p > 3$. For $Y_s = (f_s, f_s', f_s'')^{\mathsf{T}}$ one has

$$Y_s - B_\phi Y_{s-1}^\phi \in p^{3s}A^3 \qquad (s \geq 1). \tag{42}$$

Here is the proof, including the change-of-lift precision. On the ordinary localization A_H, the canonical lift σ has $h(q) \mapsto h(q^p)$. The split Tate degeneration of (37), and $D(0) = 1$, normalize the Hodge coefficient as $c = p/\alpha$ with $\alpha = a(h)/a(h^\sigma)$. Horizontality then gives

$$M_\sigma = \begin{pmatrix} c & 0 \\ c'/\sigma' & c/\sigma' \end{pmatrix}, \quad V_\sigma = \begin{pmatrix} \alpha & 0 \\ \alpha' & \alpha\sigma' \end{pmatrix}.$$

Therefore B_σ is exactly the chain-rule matrix obtained by differentiating $a(h)^2/a(h^\sigma)^2$ twice. Its three columns are divisible by $1, p, p^2$, respectively. The scalar formal congruence (36), together with its first two derivatives, gives (42) for σ on the ordinary chart. To pass from the formal cusp identity to A_H, use the q-expansion principle modulo p^e: the mod-p special fiber is integral, hence localization and expansion into the Laurent series field at $h = 0$ are injective, and successive division by p proves saturation at every precision. For the underlying ordinary canonical quotient and the normalized Tate differential, see Katz, *p-adic properties of modular schemes and modular forms*, Sections 3.1–3.4 and Appendix A2, especially (A2.2.1) and (A2.2.7) [3]. Its Proposition 2.7.1

gives the integral q-expansion principle. The equality $q_E = q^6$ above checks that these standard constructions have exactly the normalization used here.

Now $\delta = \phi(h) - \sigma(h)$ is divisible by p. Let $T(\sigma, \phi)$ send a horizontal jet at $\phi(h)$ to its horizontal continuation at $\sigma(h)$. The crystalline change-of-lift identity, with the row convention for M, is

$$B_\phi = B_\sigma T(\sigma, \phi).$$

This is the Taylor comparison of the same crystal at two lifts; see the explicit divided-power formula in Stacks Project, Proposition 60.17.4, Tag 07JH [4]. Horizontal Taylor transport is integral: its terms have the form $\delta^n/n!$ times iterated integral connection matrices, and $n - v_p(n!) \to \infty$. The residual (41) shows that

$$T(\sigma, \phi)Y^\phi_{s-1} - Y^\sigma_{s-1} \in p^{3(s-1)} \left(p^3 A_H \oplus p^2 A_H \oplus p A_H\right).$$

Indeed, solve the homogeneous initial-value problem having the same three-jet at the starting point. The difference from f_{s-1} has zero initial jet and forcing term in $p^{3(s-1)}A_H$. Successive integration starts the three coordinates at Taylor orders $3, 2, 1$. For $p > 3$ and $n \geq k$, where $k = 1, 2, 3$, one has $n - v_p(n!) \geq k$. Thus the three coordinate valuations are at least

$$3(s-1)+3, \qquad 3(s-1)+2, \qquad 3(s-1)+1.$$

Multiplication by the three columns of B_σ raises all three error components to p^{3s}. The displayed change-of-lift identity therefore gives the desired congruence for B_ϕ. This proves (42) on A_H. Both sides belong to A^3; the injection $A/p \hookrightarrow A_H/p$ makes the localization p-saturated, and yields (42) on A.

The inputs from crystalline theory in this argument are its integral Frobenius and Verschiebung on a smooth elliptic family, horizontality, and the Taylor change-of-lift identity. The normalizations, the unit change of basis, and the extra three-digit precision are established explicitly above.

6.5 The supersingular congruence at every cutoff

The CM hypothesis is unnecessary on the supersingular side. The explicit level-six family supplies a stronger statement.

Theorem 6.1 (Supersingular Apéry residue disks)**.** *Let $p > 3$ and $t_0 \in \mathbb{Z}_p$ satisfy*

$$t_0(1 - 34t_0 + t_0^2) \in \mathbb{Z}_p^\times.$$

Choose a root h_0 of

$$72t_0h^2 + (17t_0 - 1)h + t_0 = 0$$

in the ring $\mathcal{O}$ of an unramified extension of degree at most two. Assume the reduction of the explicit elliptic curve E_{h_0} in (37) is supersingular. Then, for every positive integer N divisible by p^2,

$$P_N(t_0) - p^2 P_{N/p^2}(t_0) \in p^{3v_p(N)-1}\mathbb{Z}_p. \tag{43}$$

The integral Frobenius–Taylor identity (42) was proved in the preceding section. No CM condition on t_0 occurs in the argument below.

Proof. The discriminant of the displayed quadratic is $1 - 34t_0 + t_0^2$, a unit. Its leading and constant coefficients are units, since $p > 3$ and t_0 is a unit. Thus its two roots are units in an unramified extension of degree at most two. The identity $D(h_0) = h_0/t_0$ makes $D(h_0)$ a unit as well. Consequently both the elliptic curve and its D-twist have good reduction.

The untwisted family has a rational point of exact order six:

$$R = \left(h(1+9h), \frac{hD}{2}\right), \qquad 2R = \left(h^2, \frac{h^2(1+8h)}{2}\right), \qquad 3R = (0,0).$$

These formulas follow directly from the addition law. The assumptions that hD is a unit and $p > 3$ preserve the exact order after reduction.

First suppose $h_0 \in \mathbb{Z}_p$. Its supersingular reduction is defined over $\mathbb{F}_p$. The Frobenius trace is divisible by p and has absolute value at most $2\sqrt{p} < p$ for $p > 3$, hence it is zero. This holds for its quadratic twist as well. Choose

$$\phi(h) = h_0 + (h - h_0)^p.$$

It is an integral Frobenius lift fixing h_0. On the rank-two fiber, $M^2 = -pI$, so on the jet module $B^2 = p^2 I$. The first row of M is divisible by p by the Hodge filtration, hence the first row of $V = pM^{-1} = -M$ is divisible by p, and that of B by p^2.

Next suppose h_0 has residue degree two. Write ι for the nontrivial automorphism of $\mathcal{O}/\mathbb{Z}_p$. Its two roots are h_0, h_0^ι. We may choose an integral Frobenius lift that exchanges them exactly: divide h^p by the monic quadratic minimal polynomial, let $r_p(h)$ be the remainder, let $r(h)$ be the affine map exchanging the roots, and put

$$\phi(h) = h^p - r_p(h) + r(h).$$

The irreducibility of the quadratic modulo p gives $r_p \equiv r \pmod p$, so this is indeed a Frobenius lift.

The reduction of E_{h_0} over $\mathbb{F}_{p^2}$ is supersingular. Its Frobenius trace is cp with $c \in \{-2, -1, 0, 1, 2\}$: supersingularity gives divisibility by p, and the Weil bound gives this range. Its rational point of order six implies

$$6 \mid \#E_{h_0}(\mathbb{F}_{p^2}) = p^2 + 1 - cp, \qquad cp \equiv 2 \pmod 6.$$

As $p \equiv \pm 1 \pmod 6$, the only possible value of c in the stated range is $c = 2$ when $p \equiv 1 \pmod 6$, or $c = -2$ when $p \equiv -1 \pmod 6$. The p^2-Frobenius is therefore scalar $\pm p$. Indeed its characteristic polynomial is $(T \mp p)^2$. If an elliptic endomorphism is nonzero, it is an isogeny and cannot be nilpotent; therefore the relation $(\pi \mp [p])^2 = 0$ forces $\pi = \pm[p]$ exactly. A quadratic twist over $\mathbb{F}_{p^2}$ changes at most the sign, so the same scalar assertion holds for $E_{h_0}^D$.

Let M_0, M_1 be the rank-two matrices at h_0, h_0^ι with compatible bases. Their two-step product is ϵpI for $\epsilon = \pm 1$. Hence $V_0 = \epsilon M_1$ and $V_1 = \epsilon M_0$. Since each M_i has first row divisible by p, so does each V_i. Consequently the jet matrices satisfy

$$B_0 B_1 = p^2 I, \qquad \text{the first row of } B_i \text{ belongs to } p^2\mathcal{O}^3.$$

The changes from symmetric-square coordinates to jets cancel in the product. Their first row is $(1, 0, 0)$, so they preserve the indicated first-row divisibility.

Write $N = mp^s$ with $(m, p) = 1$ and $s \geq 2$. The trace argument applies to every cutoff mp^j, and the residual identity is $LP_{mp^j} = (mp^j)^3 t^{mp^j}(-A_{mp^j} + tA_{mp^j-1})$. The proof of (42) therefore gives the same identity with Y_j formed from $P_{mp^j}(t(h))$, with error in p^{3j}. Apply it twice at the two points (the same point in residue degree one). Taking the first coordinate gives

$$P_{mp^s}(t_0) - p^2 P_{mp^{s-2}}(t_0) \in p^{3s}\mathcal{O} + p^2 p^{3(s-1)}\mathcal{O} = p^{3s-1}\mathcal{O}.$$

Both polynomials have values in $\mathbb{Z}_p$, and the extension is unramified. Intersecting with $\mathbb{Z}_p$ proves (43). The argument is unchanged for every lift of the same allowed residue class, so it covers entire supersingular residue disks. □

First level and valuation consequence. The same first-row divisibility and the $s = 1$ jet identity give

$$P_{mp}(t_0) \in p^2\mathbb{Z}_p \qquad (m \geq 1).$$

Combining this with (43) shows that for every $p \mid N$,

$$v_p(P_N(t_0)) \geq 2\left\lceil \frac{v_p(N)}{2} \right\rceil.$$

In the usual prime-power notation, for all $m \geq 1$ and $s \geq 2$,

$$P_{mp^s}(t_0) - p^2 P_{mp^{s-2}}(t_0) \in p^{3(s+v_p(m))-1}\mathbb{Z}_p.$$

The extra $3v_p(m)$ follows by writing $m = p^a m_0$ with $(m_0, p) = 1$ and applying the theorem at level $s + a$. The exponent cannot uniformly be increased from $3s - 1$ to $3s$: the exact finite calculation at $p = 7$, $t_0 = 1$, $m = 1$, $s = 2$ gives

$$P_{49}(1) - 49 \equiv 2 \cdot 7^5 \pmod{7^6}.$$

6.6 The two rational CM specializations

For $t_0 = -1$, take $h_0 = -1/6$. Then $D(h_0) = 1/6$ and the curve E_{h_0} is isomorphic over $\mathbb{Z}[1/6]$ to $y^2 = x^3 + 1$. If $p \equiv 2 \pmod 3$, the map $x \mapsto x^3$ permutes $\mathbb{F}_p$, so the trace of Frobenius is zero. Thus the theorem gives

$$\sum_{k<mp^s} (-1)^k A_k - p^2 \sum_{k<mp^{s-2}} (-1)^k A_k \in p^{3(s+v_p(m))-1}\mathbb{Z}_p$$

for every such $p > 3$, $m \geq 1$, $s \geq 2$.

For $t_0 = 1$, the cover points are

$$72h^2 + 16h + 1 = 0, \qquad h_\pm = \frac{-4 \pm \sqrt{-2}}{36}.$$

At either root, write $a_2 = (1 + 12h + 24h^2)/4$ and $b = h^3(1 + 9h)$. Direct reduction modulo the quadratic gives $a_2^2 = 8b$. Hence E_h is the quadratic twist with parameter $-a_2/4$ of

$$E_0 : \quad y^2 = x^3 - 4x^2 + 2x, \qquad \Delta(E_0) = 512.$$

Moreover $a_2 = (1 + 10h)/6$ and $\mathrm{Norm}(a_2) = 1/216$, so this twist parameter is a unit at every $p > 3$.

The quotient of E_0 by $(0, 0)$ is the degree-two isogenous curve $y^2 = x^3 + 8x^2 + 8x$, its -2 quadratic twist. Over $\mathbb{Q}(\sqrt{-2})$ this gives an endomorphism J with differential multiplier $\sqrt{-2}$, and hence $J^2 = [-2]$. Its Galois conjugate is $-J$. At a prime with $(-2/p) = -1$, Frobenius anticommutes with J; therefore it has trace zero and E_0 is supersingular. The unit twists remain supersingular. This proves

$$\sum_{k<mp^s} A_k - p^2 \sum_{k<mp^{s-2}} A_k \in p^{3(s+v_p(m))-1}\mathbb{Z}_p$$

for all $p \equiv 5, 7 \pmod 8$, $m \geq 1$, and $s \geq 2$. The argument includes $p = 5, 7$: it uses the characteristic-zero explicit unit twist, so it does not incorrectly exclude primes at which the reduction of $j = 8000$ becomes 0 or 1728.

6.7 A finite supersingularity check

The theorem's hypothesis can be tested directly from its displayed elliptic curve, without an unproved identification with a truncated Apéry Hasse polynomial. Reduce h_0 to $\bar{h}$ in $\mathbb{F}_{p^2}$ (or $\mathbb{F}_p$ when possible), and form

$$\mathcal{H}_p(\bar{h}) = [X^{p-1}]\left(X^3 + \frac{1+12\bar{h}+24\bar{h}^2}{4}X^2 + \bar{h}^3(1+9\bar{h})X\right)^{(p-1)/2}.$$

The standard Cartier–Manin formula for a smooth elliptic curve says that it is supersingular exactly when $\mathcal{H}_p(\bar{h}) = 0$. The extra D-twist multiplies this coefficient by the nonzero factor $D(\bar{h})^{(p-1)/2}$ and does not change its vanishing.

6.8 Sources and scope

The modular parametrizations used above are given in Beukers, *Supercongruences using modular forms*, arXiv:2403.03301v3, Appendix Sections 8.10 and 9.5 [7]. Its Corollary 1.20 is a first-level ordinary Apéry sum congruence. Sun–Ye, arXiv:2408.09776v4, Theorem 1.29 and Section 9 prove the first-level ordinary alternating congruence and the two CM-value identities [5]. Bloch–Vanhove, arXiv:1309.5865, Section 4 gives the equivalent sunset elliptic pencil [6]. The present supersingular result covers all positive cutoffs divisible by p^2 throughout the stated smooth supersingular disks. It excludes $p = 2, 3$, $t_0 \equiv 0$, and the two branch values modulo p; no assertion about these excluded loci is made here.

Precise limits of the statements. The supersingular theorem is relative to the open set $t_0(1 - 34t_0 + t_0^2) \in \mathbb{Z}_p^\times$ and the explicit elliptic supersingularity condition. It makes no assertion at $p = 3$ or at the excluded singular reductions. The two rational specializations are corollaries of the whole-disk theorem, rather than the only parameter values covered. The modulus $3s - 1$ cannot uniformly be replaced by $3s$, as the displayed exact $p = 7$ example shows.

A Reproducible finite checks

These checks are supplementary diagnostics; no finite computation is used in the proof. A block-check script tested every odd prime $p \le 43$, every $2 \le d \le 2p + 2$, both signs satisfying $\chi = 1$, and $m = 1, 2, 3$. It found no failures in 2583 instances of (2), including nonunit cases. An independently run check tested ordinary cases with $2 \le d \le 12$ and both signs for $p = 3, 5, 7$ through level $s = 3$, and for $p = 11, 13, 17, 19, 29, 37$ through level $s = 2$. All 178 level-recursion instances passed.

The following complete Python 3 listing reproduces both counts and checks. It computes central binomial coefficients by exact integer recurrence before reducing modulo p^2. It requires no external package, data file, or approximation to the unit root.

```
def primes(bound):
    return [p for p in range(3, bound + 1, 2)
            if all(p % j for j in range(3, int(p**0.5) + 1, 2))]

def coefficients(length, modulus):
    b, inv4, q, out = 1, pow(4, -1, modulus), 1, []
    for k in range(length):
        out.append((b % modulus) * q % modulus)
        b = b * 2 * (2*k + 1) // (k + 1)
        q = q * inv4 % modulus
    return out

def terms(a, d, eps, modulus):
```

```
    return [pow(x, d, modulus) * (eps if k % 2 else 1) % modulus
            for k, x in enumerate(a)]

block_checks = 0
for p in primes(43):
    modulus, N = p*p, (p-1)//2
    a = coefficients(3*p, modulus)
    for d in range(2, 2*p + 3):
        for eps in (-1, 1):
            if (-1)**(d*N) * eps**N != 1:
                continue
            c = terms(a, d, eps, modulus)
            fp = sum(c[:p]) % modulus
            for m in (1, 2, 3):
                assert (sum(c[:m*p]) - fp*sum(c[:m])) % modulus == 0
                block_checks += 1

level_checks = 0
for p in (3, 5, 7, 11, 13, 17, 19, 29, 37):
    modulus = p*p
    max_level = 3 if p <= 7 else 2
    a = coefficients(p**max_level, modulus)
    for d in range(2, 13):
        for eps in (-1, 1):
            c = terms(a, d, eps, modulus)
            values = [sum(c[:p**s]) % modulus
                      for s in range(1, max_level + 1)]
            if values[0] % p == 0:
                continue
            for s in range(1, len(values)):
                assert (values[s] - values[0]*values[s-1]) % modulus == 0
                level_checks += 1

assert block_checks == 2583
assert level_checks == 178
print("Block checks:", block_checks, "; level checks:", level_checks)
```

B Diagnostics for the level-dependent bound

An exact modular calculation tested

$$F_{p^{s+1}}(\epsilon)F_{p^{s-1}}(\epsilon) - F_{p^s}(\epsilon)^2 \equiv 0 \pmod{p^{\nu_d(s)}}$$

in ordinary cases with $2 \le d \le 12$ and both signs. The tested primes and maximum values of s were $(3,5),(5,4),(7,3),(11,2),(13,2),(17,2)$, respectively. All 256 instances passed. These finite checks are supplementary and are not used in any proof above. They do not test optimality of the exponent. Using the `coefficients` and `terms` functions in the verification appendix, the additional check is reproduced by:

```
checks = 0
for p, maxs in ((3,5), (5,4), (7,3), (11,2), (13,2), (17,2)):
    modulus = p**(2*maxs + 1)
    a = coefficients(p**(maxs + 1), modulus)
    for d in range(2, 13):
        for eps in (-1, 1):
            c = terms(a, d, eps, modulus)
            f = [1] + [sum(c[:p**j]) % modulus
                       for j in range(1, maxs + 2)]
            if f[1] % p == 0:
                continue
            for s in range(1, maxs + 1):
                exponent = min(2*s, s+d-1)
```

```
                assert (f[s+1]*f[s-1] - f[s]**2) % p**exponent == 0
                checks += 1
assert checks == 256
print("Level-dependent cross-difference checks:", checks)
```

C Exact checks on supersingular Apéry disks

The following standalone Python 3 program selects the residue classes using the Cartier–Manin coefficient of the displayed elliptic curve over $\mathbb{F}_{p^2}$. It does not select them by assuming the Apéry congruence being tested. It checks all the selected smooth residue classes for the listed primes through 43, at lifts t and $t+p$, and also higher levels at several lifts. All 94 exact checks pass. The finite computation is a diagnostic for the theorem, not its proof.

```
#!/usr/bin/env python3
"""Exact diagnostics for the supersingular Apery-disk theorem.

Supersingularity is tested on the displayed elliptic curve over F_(p^2),
not inferred from the Apery truncation being tested. No floating point.
"""
from math import comb


def add(x, y, p):
    return ((x[0]+y[0]) % p, (x[1]+y[1]) % p)


def scale(x, c, p):
    return (x[0]*c % p, x[1]*c % p)


def mul(x, y, p, d):
    return ((x[0]*y[0]+d*x[1]*y[1]) % p,
            (x[0]*y[1]+x[1]*y[0]) % p)


def power(x, k, p, d):
    ans = (1, 0)
    while k:
        if k & 1:
            ans = mul(ans, x, p, d)
        x = mul(x, x, p, d)
        k >>= 1
    return ans


def ss_residues(p):
    d = next(i for i in range(2, p) if pow(i, (p-1)//2, p) == p-1)
    roots = []
    for t in range(1, p):
        disc = (1-34*t+t*t) % p
        if not disc:
            continue
        if pow(disc, (p-1)//2, p) == 1:
            root = (next(i for i in range(p) if i*i % p == disc), 0)
        else:
            ratio = disc * pow(d, -1, p) % p
            root = (0, next(i for i in range(p) if i*i % p == ratio))
        h = scale(add((1-17*t, 0), root, p), pow(144*t, -1, p), p)
        h2 = mul(h, h, p, d)
        a = scale(add(add((1, 0), scale(h, 12, p), p), scale(h2, 24, p), p),
                  pow(4, -1, p), p)
        b = mul(power(h, 3, p, d), add((1, 0), scale(h, 9, p), p), p, d)
        m = (p-1)//2
        hasse = (0, 0)
        for j in range(m//2+1):
            term = mul(power(a, m-2*j, p, d), power(b, j, p, d), p, d)
            hasse = add(hasse, scale(term, comb(m, j)*comb(m-j, j), p), p)
```

```
        if hasse == (0, 0):
            roots.append(t)
    return roots


def apery_coefficients(length, modulus):
    last, a = 0, 1
    result = []
    for n in range(length):
        result.append(a % modulus)
        numerator = (2*n+1)*(17*n*(n+1)+5)*a-n**3*last
        denominator = (n+1)**3
        assert numerator % denominator == 0
        last, a = a, numerator//denominator
    return result


def evaluate(coefficients, t, modulus):
    value = 0
    for c in reversed(coefficients):
        value = (value*t+c) % modulus
    return value


def valuation_residue(value, p, precision):
    value %= p**precision
    if value == 0:
        return f">={precision}"
    v = 0
    while value % p == 0:
        value //= p
        v += 1
    return v


def main():
    rows = []
    precision = 10
    for p in (5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43):
        modulus = p**precision
        coeffs = apery_coefficients(p*p, modulus)
        residues = ss_residues(p)
        for residue in residues:
            for t in (residue, residue+p):
                defect = evaluate(coeffs, t, modulus)-p*p
                val = valuation_residue(defect, p, precision)
                assert isinstance(val, str) or val >= 5
                rows.append(dict(p=p, t=t, s=2, m=1, defect_valuation=val))
        print(p, residues, flush=True)
    for p, t, depth in ((5, 6, 4), (5, 11, 4), (5, 4, 4),
                        (5, 9, 4), (7, 8, 3), (7, 15, 3),
                        (11, 10, 3), (11, 21, 3)):
        assert t % p in ss_residues(p)
        precision = 3*depth+4
        modulus = p**precision
        coeffs = apery_coefficients(p**depth, modulus)
        vals = {s:evaluate(coeffs[:p**s], t, modulus) for s in range(depth+1)}
        for s in range(2, depth+1):
            val = valuation_residue(vals[s]-p*p*vals[s-2], p, precision)
            assert isinstance(val, str) or val >= 3*s-1
            rows.append(dict(p=p, t=t, s=s, m=1, defect_valuation=val))
    assert len(rows) == 94
    print(f'{len(rows)} exact supersingular-disk checks passed')


if __name__ == '__main__':
    main()
```

PART II

All-Level Beukers–Delaygue Supercongruences via Lifted Reflection

Research manuscript prepared for Keio

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Abstract

Let $F_L(\epsilon) = \sum_{n<L}((1/2)_n/n!)^d\epsilon^n$, where $d \geq 2$ and $\epsilon = \pm 1$. For every odd prime p for which $F_p(\epsilon)$ is a p-adic unit, we prove

$$F_{mp^s}(\epsilon) - f_p(\epsilon)F_{mp^{s-1}}(\epsilon) \in p^{2s}\mathbb{Z}_p \qquad (m, s \geq 1).$$

Here $f_p(\epsilon)$ is the limiting quotient of successive prime-power truncations; the proof also constructs this limit. At $m = 1$ this is the full statement of Beukers–Delaygue's Conjecture 1.5. The proof treats both ranges $s < d$ and $s \geq d$ without a restriction on the size of p. It identifies the carry contribution in the analytic digit operator as an exact weighted finite difference. A two-component integral lift then has rank at most one modulo p^2, which gives a two-digit gain at every normalized iteration. All analytic interpolation and operator identities are proved here; no unpublished supercongruence theorem is assumed.

Scope and research status. The theorem covers every level of Conjecture 1.5 under its original unit hypothesis, including $p = 3$. It also allows every positive truncation multiplier m, without assuming $F_m(\epsilon)$ is a unit. The proof has received internal independent checks. This manuscript does not claim completed external peer review or certified literature priority. The stronger special exponents and modular identifications in Conjectures 1.3 and 1.4 are separate statements.

1 Statement and proof strategy

Fix an odd prime p, an integer $d \geq 2$, and $\epsilon \in \{1, -1\}$. Write

$$\alpha_n = \frac{(1/2)_n}{n!} = 4^{-n}\binom{2n}{n}, \qquad F_L(\epsilon) = \sum_{n=0}^{L-1} \alpha_n^d \epsilon^n, \qquad N = \frac{p-1}{2}.$$

All coefficients belong to $\mathbb{Z}_p$. The following theorem proves the mathematical statement of Beukers–Delaygue's Conjecture 1.5 [1], with the additional allowance of arbitrary truncation multipliers.

Theorem 1.1 (All levels and all lengths)**.** *Assume $F_p(\epsilon) \in \mathbb{Z}_p^\times$. The limit*

$$f_p(\epsilon) = \lim_{s\to\infty} \frac{F_{p^s}(\epsilon)}{F_{p^{s-1}}(\epsilon)}$$

exists in $\mathbb{Z}_p^\times$. For all integers $m, s \geq 1$,

$$F_{mp^s}(\epsilon) - f_p(\epsilon)F_{mp^{s-1}}(\epsilon) \in p^{2s}\mathbb{Z}_p. \tag{1}$$

There is no unit condition on $F_m(\epsilon)$. Equivalently, for every positive integer K divisible by p,

$$F_K(\epsilon) - f_p(\epsilon)F_{K/p}(\epsilon) \in p^{2v_p(K)}\mathbb{Z}_p. \tag{2}$$

The theorem includes $p = 3$ and imposes no condition relating p and d.

The proof replaces a direct parity estimate by an operator on a larger integral space. An exact finite-difference term records the failure of the full digit operator to commute with reflection. The larger space retains that term, and its operator contracts every normalized nonconstant component by p^2 at each iteration. No inverse of a finite-difference operator, and no assertion that its image is closed, is required.

2 Integral analytic digit functions

Let $\mathcal{A} = \mathbb{Z}_p\langle a\rangle$ be the algebra of restricted power series, with its Gauss norm. Divisibility in this algebra always means coefficientwise divisibility. Integral affine substitutions and evaluations at points of $\mathbb{Z}_p$ have norm at most one.

Lemma 2.1. *There are functions $g_t \in \mathcal{A}$, $0 \le t < p$, satisfying*

$$g_t(a) = \frac{\alpha_{pa+t}}{\alpha_a} \quad \textit{for every nonnegative integer } a.$$

For $0 \le t \le N$, there is a $\beta_t \in \mathbb{Z}_p$ such that

$$g_t(a) \equiv \alpha_t(1 + p\beta_t a) \pmod{p^2\mathcal{A}}.$$

In particular its constant term is α_t and $g_t \equiv \alpha_t \pmod{p\mathcal{A}}$. For $N < t < p$,

$$g_t(a) = p(a + 1/2)b_t(a), \qquad b_t \in \mathcal{A}. \tag{3}$$

Proof. Cancellation of factors divisible by p gives, initially at nonnegative integer a,

$$\frac{g_0(a+1)}{g_0(a)} = C\frac{\prod_{0\le t<p,\ t\ne N}(1 + pa/(t+1/2))}{\prod_{i=1}^{p-1}(1+pa/i)}, \qquad C = \frac{\prod_{0\le t<p,\ t\ne N}(t+1/2)}{(p-1)!}.$$

Both lists of constant unit factors reduce to $\mathbb{F}_p^\times$, so $C \in 1 + p\mathbb{Z}_p$ and, for $j \ge 1$,

$$s_j = \sum_{0\le t<p,\ t\ne N} (t+1/2)^{-j} - \sum_{i=1}^{p-1} i^{-j} \in p\mathbb{Z}_p.$$

Define the polynomial

$$P_j(a) = \sum_{v=0}^{j} \left\{ {j \atop v} \right\} \frac{a(a-1)\cdots(a-v)}{v+1}.$$

The braces are Stirling numbers of the second kind. Thus $P_j(a) = \sum_{u=0}^{a-1} u^j$ at nonnegative integers, and every coefficient of P_j has valuation at least $-\lfloor \log_p(j+1) \rfloor$. The series

$$L(a) = a\log C + \sum_{j\ge1} \frac{(-1)^{j-1}p^j s_j}{j} P_j(a)$$

converges in $\mathcal{A}$, since its jth nonconstant summand has coefficient valuations at least

$$j + 1 - v_p(j) - \lfloor \log_p(j+1) \rfloor \ge 2,$$

and these bounds tend to infinity. The inequality is immediate for $j = 1, 2, 3$; for $j \ge 4$ it follows from $p \ge 3$ and $3^{j-1} \ge j(j+1)$. In particular $L \in p\mathcal{A}$ and $L - a\log C \in p^2\mathcal{A}$. Set $g_0 = \exp L$.

Its difference equation and value $g_0(0) = 1$ give the required integer values, and $g_0 \equiv 1 + a \log C \pmod{p^2\mathcal{A}}$.

For $1 \le t < p$, set

$$g_t(a) = g_0(a)4^{-t}\frac{\prod_{j=1}^{2t}(2pa+j)}{\prod_{j=1}^{t}(pa+j)^2}.$$

All denominator constants are units. This product proves both the integer-value assertion and the affine congruence when $t \le N$. When $t > N$, its numerator contains the factor $2pa+p = 2p(a+1/2)$; cancelling this factor gives (3) in $\mathcal{A}$. □

3 Both exact reflections

Let $r(a) = -a - 1/2$ and $r_1(a) = -a - 3/2$.

Lemma 3.1. *The digit functions satisfy the following exact identities in $\mathcal{A}$:*

$$g_{N-t}(r(a)) = (-1)^N g_t(a) \qquad (0 \le t \le N), \tag{4}$$

$$b_{p+N-t}(r_1(a)) = (-1)^{N+1} b_t(a) \qquad (N < t < p). \tag{5}$$

Proof. Put $N_j = (p^j - 1)/2$. For each fixed nonnegative integer n,

$$\frac{\alpha_{N_j-n}}{\alpha_{N_j}} = \prod_{k=0}^{n-1}\frac{N_j-k}{N_j-k-1/2} \longrightarrow \alpha_n.$$

Also

$$\frac{\alpha_{N_j}}{\alpha_{N_{j-1}}} = g_N(N_{j-1}) \longrightarrow C_* := g_N(-1/2).$$

For fixed $a \ge 0$ and $0 \le t \le N$, use $a_j = N_{j-1} - a$ and $n = pa + t$. The equality $pa_j + N - t = N_j - n$ and the preceding limits give

$$g_{N-t}(r(a)) = C_* g_t(a).$$

Continuity extends this equality to $\mathbb{Z}_p$, hence to an identity of restricted power series. Applying it twice gives $C_*^2 = 1$. Since $g_N \equiv \alpha_N \equiv (-1)^N \pmod{p}$, one has $C_* = (-1)^N$. This proves (4).

For $N < t < p$, let $t^* = p + N - t$, which also lies strictly between N and p. Now use $a_j = N_{j-1} - a - 1$, so that $pa_j + t^* = N_j - (pa + t)$. The same finite-product limits give

$$g_{t^*}(r_1(a)) = (-1)^N\frac{\alpha_{pa+t}}{\alpha_{a+1}} = (-1)^N\frac{a+1}{a+1/2}g_t(a) \qquad (a \ge 0).$$

Using (3) on both sides cancels the displayed factor and gives (5). This last identity is between elements of $\mathcal{A}$, so continuity proves it everywhere. □

Put $\chi = (-1)^{dN}\epsilon^N$. The ordinary assumption forces

$$\chi = 1. \tag{6}$$

Indeed $\alpha_t \equiv (-1)^t\binom{N}{t} \pmod{p}$ for $0 \le t \le N$, so $\alpha_{N-t} \equiv (-1)^N\alpha_t \pmod{p}$. The other digits are divisible by p. Reflection of the lower-digit sum therefore gives $F_p(\epsilon) \equiv \chi F_p(\epsilon) \pmod{p}$. Since p is odd and the sum is a unit, (6) follows.

4 The finite-difference identity and exact symmetrization

Define bounded integral operators on $\mathcal{A}$ by

$$(Sf)(a) = \sum_{t=0}^{p-1} \epsilon^t g_t(a)^d f(pa+t), \qquad (Rf)(a) = f(-a-1/2),$$

and

$$(Df)(a) = a^d f(a) - \epsilon(a+1/2)^d f(a+1), \qquad (Uf)(a) = g_0(a)^d f(pa). \tag{7}$$

Let

$$(Bf)(a) = \sum_{t=N+1}^{p-1} \epsilon^t b_t(a)^d f(pa+t), \qquad (E_- f)(a) = f(a-1).$$

The upper-digit part of S is consequently $H = p^d(a+1/2)^d B$; all these operators are bounded and integral.

Lemma 4.1 (Two exact operator identities)**.** *One has*

$$SD = p^d DU, \tag{8}$$

$$RSR - S = \epsilon p^d DE_- B. \tag{9}$$

Consequently, with

$$T = \frac{S + RSR}{2}, \qquad V = -\frac{\epsilon}{2} E_- B, \tag{10}$$

the operator T commutes with R and

$$S = T + p^d DV. \tag{11}$$

Proof. The summands of SDf telescope by the recurrence $\alpha_{n+1}/\alpha_n = (n+1/2)/(n+1)$. The two surviving terms are

$$p^d a^d g_0(a)^d f(pa) - \epsilon p^d (a+1/2)^d g_0(a+1)^d f(p(a+1)),$$

which proves (8). This may first be checked at nonnegative integers; the displayed functions all belong to $\mathcal{A}$, so the identity holds there. No division by $a+1$ is needed in the final identity.

The lower-digit operator commutes with R, by (4), (6), and $r(pa+t) = pr(a) + (N-t)$. For the upper digits, (5) and $pr_1(a) + (p+N-t) = r(pa+t)$ give

$$(Bf)(r_1(a)) = \epsilon(-1)^d (BRf)(a).$$

It follows that

$$(RHf)(a) = \epsilon p^d a^d (BRf)(a-1), \qquad (HRf)(a) = p^d (a+1/2)^d (BRf)(a).$$

Hence $RS - SR = \epsilon p^d DE_- BR$. Multiplication on the right by R, with $R^2 = 1$, proves (9). The remaining assertions follow immediately. All divisions used in (10) are by the unit 2. □

The finite-difference operator has a useful interpretation. For

$$\mathcal{L}_m(f) = \sum_{a=0}^{m-1} \alpha_a^d \epsilon^a f(a),$$

direct telescoping gives

$$\mathcal{L}_m(Df) = -m^d \alpha_m^d \epsilon^m f(m).$$

In particular this boundary term is divisible by p^{ds} when $m = p^s$. The proof below retains Df in an integral state space instead of discarding it at any finite stage.

5 An integral lift with two-digit contraction

Let $\mathcal{A}^+ = \{e \in \mathcal{A} : Re = e\}$, and set

$$\mathcal{M} = \mathcal{A}^+ \oplus \mathcal{A}, \qquad q(e, b) = e + Db.$$

Equip this complete $\mathbb{Z}_p$-module with the maximum Gauss norm. Define

$$\mathcal{K}(e, b) = (Te,\ p^d Ve + p^d Ub). \tag{12}$$

It is a bounded integral endomorphism because T preserves $\mathcal{A}^+$. Equations (8) and (11) give the exact intertwining identity

$$q\mathcal{K} = Sq. \tag{13}$$

Put $c = -1/4$, $v_0 = (1, 0)$, and $\ell(e, b) = e(c)$. Then $\ell(v_0) = 1$, and $\mathcal{M} = \mathbb{Z}_p v_0 \oplus \ker \ell$. The crucial improvement is

$$(1 - v_0\ell)\mathcal{K}(\mathcal{M}) \subseteq p^2 \ker \ell, \qquad \ell(\mathcal{K}v_0) \in \mathbb{Z}_p^\times. \tag{14}$$

To see the first assertion, Lemma 2.1 and $f(pa + t) \equiv f(t) + paf'(t) \pmod{p^2\mathcal{A}}$ show that Sf is affine modulo p^2 for every $f \in \mathcal{A}$; the upper digits vanish modulo p^2 because $d \geq 2$. Since R is an integral affine substitution, $RSRf$ is also affine modulo p^2. For $e \in \mathcal{A}^+$, Te is even under $a \mapsto -a - 1/2$. An affine even function modulo p^2 is constant modulo p^2, so $Te - Te(c) \in p^2\mathcal{A}^+$. The second component of (12) is divisible by p^d, hence by p^2. Finally $\ell(\mathcal{K}v_0) \equiv F_p(\epsilon) \pmod{p}$, which proves the unit assertion.

Lemma 5.1 (Normalized iteration)**.** *There is a unit $\lambda \in \mathbb{Z}_p^\times$ such that*

$$\mathcal{K}^s v_0 - \lambda \mathcal{K}^{s-1} v_0 \in p^{2s}\mathcal{M} \qquad (s \geq 1). \tag{15}$$

Proof. On $\mathcal{Q} = v_0 + p^2 \ker \ell$ define $\mathcal{N}(x) = \mathcal{K}x/\ell(\mathcal{K}x)$. By (14), the denominator is a unit and $\mathcal{N}(\mathcal{Q}) \subseteq \mathcal{Q}$. For $x, y \in \mathcal{Q}$ one has the exact identity

$$\mathcal{N}(x) - \mathcal{N}(y) = \frac{\mathcal{K}(x - y) - \ell(\mathcal{K}(x - y))\mathcal{N}(y)}{\ell(\mathcal{K}x)}.$$

If $x - y \in p^r \ker \ell$, then the numerator equals

$$(1 - v_0\ell)\mathcal{K}(x - y) - \ell(\mathcal{K}(x - y))(\mathcal{N}(y) - v_0),$$

which belongs to $p^{r+2} \ker \ell$ by (14). Thus normalized iteration gains two powers of p at every step.

Set $h_0 = v_0$ and $h_s = \mathcal{N}(h_{s-1})$. Then $h_s - h_{s-1} \in p^{2s} \ker \ell$, so h_s converges to $h_\infty \in \mathcal{Q}$. Write

$$Z_s = \ell(\mathcal{K}^s v_0), \quad z_s = Z_s/Z_{s-1} = \ell(\mathcal{K}h_{s-1}), \quad \lambda = \ell(\mathcal{K}h_\infty).$$

Inductively Z_s and z_s are units and $h_s = \mathcal{K}^s v_0/Z_s$. Since $h_\infty - h_{s-1} \in p^{2s}\mathcal{M}$, one has $z_s - \lambda \in p^{2s}\mathbb{Z}_p$. The identity

$$\mathcal{K}^s v_0 - \lambda \mathcal{K}^{s-1} v_0 = Z_{s-1}\big(z_s(h_s - h_{s-1}) + (z_s - \lambda)h_{s-1}\big)$$

now proves (15). □

6 Return to the original truncation lengths

Applying the bounded integral map q to (15), and using $q(v_0) = 1$ and (13), gives the uniform function congruence

$$S^s 1 - \lambda S^{s-1} 1 \in p^{2s}\mathcal{A}. \tag{16}$$

The exact digit expansion gives, for every $m \geq 1$ and $s \geq 0$,

$$\mathcal{L}_m(S^s 1) = F_{mp^s}(\epsilon). \tag{17}$$

Indeed an iterated digit sequence represents a unique $0 \leq b < p^s$; the product of the g_t factors telescopes to $\alpha_{p^s a+b}/\alpha_a$. Since p is odd, the accompanying sign after multiplication by ϵ^a is $\epsilon^{p^s a+b}$. This is an exact rational identity and needs no unit assumption on α_a or $F_m(\epsilon)$.

For $m = 1$, $F_{p^s}(\epsilon) = (S^s 1)(0)$. Since $h_s \in v_0 + p^2 \ker \ell$ and q is integral, $q(h_s)(0) \in 1 + p^2\mathbb{Z}_p$. Hence these truncations are units, and (16) implies

$$\frac{F_{p^s}(\epsilon)}{F_{p^{s-1}}(\epsilon)} \equiv \lambda \pmod{p^{2s}}.$$

This proves existence of the stated limit and identifies it with $\lambda = f_p(\epsilon)$. Finally apply the bounded integral functional $\mathcal{L}_m$ to (16) to obtain (1). Writing $K = mp^{v_p(K)}$ with $p \nmid m$ yields (2) and completes the proof of Theorem 1.1.

Why the earlier parity estimate lost precision. In the original function space the upper digits feed a reflection-odd component of size p^d, whose naive estimate subsequently gains only one digit per iteration. Identity (9) identifies this contribution as an exact D term. The lifted space stores its primitive as the second coordinate; (8) makes that coordinate gain $d \geq 2$ digits at every iteration. The argument therefore removes the obstruction rather than assuming its value vanishes at the evaluation point $a = 0$.

7 Consequences and precise scope

Taking $m = 1$ in Theorem 1.1 gives the complete statement of Conjecture 1.5 of [1]. Taking $s = 1$ gives Conjecture 1.2, since $F_1(\epsilon) = 1$. In particular, the range $s \geq d$ left open by the earlier direct parity estimate is included in the same induction as the smaller levels.

Parameter or condition	Range in this manuscript
Prime	Every odd prime, including 3
Number of parameters	Every integer $d \geq 2$
Sign	$\epsilon \in \{1, -1\}$
Ordinary hypothesis	$F_p(\epsilon) \in \mathbb{Z}_p^\times$
Level	Every $s \geq 1$, including $s \geq d$
Truncation multiplier	Every $m \geq 1$; $F_m(\epsilon)$ may be a nonunit
Proved exponent	$2s$, or $2v_p(K)$ for a cutoff K divisible by p

The unit hypothesis is exactly the one in the original conjecture. The theorem makes no assertion for $p = 2$, for other values of ϵ, or for nonunit $F_p(\epsilon)$. Neither the larger special exponents in Conjecture 1.3 nor the modular-coefficient identifications in Conjecture 1.4 follow from Theorem 1.1 alone.

The earlier direct parity bound $\min(2s, s+d-1)$ records the upper-digit contribution only by its size. The operator identity (8) records additional exact structure: the primitive of every carry correction acquires p^d at its next iteration. Keeping that primitive as a separate coordinate removes the artificial one-digit loss. This is an argument for every level, not a finite verification followed by extrapolation.

A Reproducible exact diagnostics

The proof does not depend on the following computations. They test the two necessary finite identities

$$F_{p^{s+1}}F_{p^{s-1}} - F_{p^s}^2 \equiv 0 \pmod{p^{2s}}$$

and

$$F_{mp^s}F_{p^{s-1}} - F_{p^s}F_{mp^{s-1}} \equiv 0 \pmod{p^{2s}}, \qquad m = 2, 3, 4,$$

with a common sign ϵ in every factor. We use the prime and maximum-level pairs

$$(p, s_{\max}) = (3, 8),\ (5, 6),\ (7, 4),\ (11, 3),$$

with $2 \le d \le 8$ and both signs whenever $F_p(\epsilon)$ is a unit. All 656 checks pass. Of these checks, 236 are in the previously uncovered range $s \ge d$.

The program uses exact modular arithmetic. In updating α_n it separates powers of p before inverting denominator units, so no division by a nonunit modulo p^k is performed. The recorded valuations are lower bounds when a residue is zero at the working precision.

```
from time import perf_counter

def coeffs(p, maxn, prec):
    mod=p**prec
    u,v=1,0
    out=[]
    for n in range(maxn):
        out.append(u*pow(p,v,mod)%mod if v<prec else 0)
        a,b=2*n+1,2*n+2
        while a%p==0:
            a//=p;v+=1
        while b%p==0:
            b//=p;v-=1
        u=u*a*pow(b,-1,mod)%mod
        assert v>=0
    return out

def vp(n,p,prec):
    if n==0:return prec
    ans=0
    while n%p==0:ans+=1;n//=p
    return ans

start=perf_counter()
checks=0
counts=[]
for p,maxs in ((3,8),(5,6),(7,4),(11,3)):
    prec=2*maxs+3
    mod=p**prec
    largest=max(p**(maxs+1),4*p**maxs)
    a=coeffs(p,largest,prec)
    for d in range(2,9):
        for eps in (-1,1):
            prefix=[0]
            for n,x in enumerate(a):
```

```
                prefix.append((prefix[-1]+pow(x,d,mod)*(eps if n%2 else 1))%mod)
            if prefix[p]%p==0:continue
            vals=[prefix[p**s] for s in range(maxs+2)]
            min_excess=prec
            for s in range(1,maxs+1):
                defect=(vals[s+1]*vals[s-1]-vals[s]**2)%mod
                assert defect%p**(2*s)==0,(p,d,eps,s,'cross',vp(defect,p,prec))
                checks+=1
                min_excess=min(min_excess,vp(defect,p,prec)-2*s)
                for m in (2,3,4):
                    defect=(prefix[m*p**s]*vals[s-1]-vals[s]*prefix[m*p**(s-1)])%mod
                    assert defect%p**(2*s)==0,(p,d,eps,s,m,vp(defect,p,prec))
                    checks+=1
            counts.append((p,d,eps,maxs,min_excess))
    print('prime',p,'done',flush=True)
print('all exact checks passed:',checks)
print('seconds:',round(perf_counter()-start,2))
print('metadata (p,d,eps,maxs,min consecutive excess):',counts)

assert checks == 656
previously_uncovered = sum(4*max(0, maxs-d+1)
                           for p,d,eps,maxs,excess in counts)
assert previously_uncovered == 236
print("Checks in the former s >= d gap:", previously_uncovered)
```

PART III

The AESZ4 Attractor and Its Residual Monodromy

QEEDDD Verifiable-Mathematical-Reasoning-Corpus Project

10 November 2026, revised for this edition

Abstract

For the fourth-order hypergeometric system AESZ4, we compute the exact Picard–Fuchs error of every truncation and classify the residual monodromy of the standard companion lattice at every auxiliary prime. The geometric action is irreducible modulo $\ell \neq 3$. Modulo 3 it is a single regular unipotent block, with a unique invariant flag. A modular-polynomial calculation excludes a generic fiberwise 2-isogeny between the two elliptic factors. We record the ordinary unit-root criterion that motivated the search for a rank-two summand, including the integral comparison that this criterion requires. Part IV constructs the summand at the attractor and proves the special-value congruence for every $p \geq 5$.

1 The two Frobenius comparisons

Let V_{z_*} denote the rank-four constituent of the third cohomology selected by the AESZ4 family. Its middle cohomology in the Hesse fiber-product model has dimension 40. Two comparisons enter the proof of Beukers's congruence. The first identifies a rank-two summand of V_{z_*}, its holomorphic line, and its Frobenius polynomial $X^2 - a_p X + p^3$. The second connects the action of Frobenius on this line to the finite hypergeometric sum modulo p^3.

Theorem 4.1 below controls the differential equation at the cutoff. Theorem 7.1 classifies the residual monodromy of the varying family. At $z_* = -1/5832$ there are additional symmetries of the special fiber. Part IV, Appendix A, uses them to construct the required summand, and its Lemma 4.1 supplies the integral comparison.

2 The conjecture, the family, and normalizations

Define

$$F(t) = {}_4F_3\left(\begin{matrix}\frac{1}{3},\frac{1}{3},\frac{2}{3},\frac{2}{3}\\ 1,1,1\end{matrix};t\right) = \sum_{n\geq 0} c_n t^n, \qquad c_n = \frac{(1/3)_n^2 (2/3)_n^2}{(n!)^4}.$$

For an integer $N \geq 1$ write

$$P_N(t) = \sum_{n=0}^{N-1} c_n t^n.$$

Because

$$\binom{-1/3}{k}^2 \binom{-2/3}{k}^2 = \frac{(1/3)_k^2 (2/3)_k^2}{(k!)^4},$$

Beukers's sum is exactly $P_p(-1/8)$.

Conjecture 2.1 (Good-prime form of Beukers, Conjecture 1.28)**.** *For every odd prime $p \neq 3$,*

$$P_p\left(-\frac{1}{8}\right) \equiv a_p \pmod{p^3},$$

where a_p is the coefficient of q^p in the rational newform

$$f_{54}(q) = q + 2q^2 + 4q^4 + 3q^5 + 29q^7 + 8q^8 + \cdots$$

of weight 4 *and level* 54*, LMFDB label* 54.4.*a.c.*

The printed statement in [7, Conjecture 1.28] has $p \neq 3$. At $p = 2$ the sum is $161/162$, whereas $a_2 = 2$. Their difference has 2-adic valuation -1. The good-prime statement above is proved in Part IV, Theorem 1.1.

The hypergeometric coordinate is related to the mirror-bicubic coordinate z by

$$t = 729z.$$

Thus the attractor point

$$z_* = -\frac{1}{5832}$$

corresponds to

$$t_* = -\frac{1}{8}.$$

The modular decomposition of the rank-four constituent uses the weight-4 form 54.4.*a.c* and the weight-2 form 54.2.*a.a*.

3 Excellent lifts and specialization

Coefficientwise congruences of Dwork type motivated the original ordinary approach [8, 9]. For example, the formal descent one might seek for the Hadamard square of the cubic simplicial period has the shape

$$\frac{P_{mp^s}(t)}{F(t)} \equiv \frac{P_{mp^{s-1}}(\sigma(t))}{F(\sigma(t))} \pmod{p^{2s}}.$$

Here $m, s \geq 1$ and σ denotes a prospective excellent Frobenius lift. Using such an identity at t_* would require an analytic ring admitting that specialization, a comparison of t_* with $\sigma(t_*)$, and an arithmetic interpretation of the multiplier. At $m = s = 1$ the displayed exponent is two, whereas Beukers asks for three. Part IV obtains the required precision directly from factorial blocks and the rational lift z_* of the special fiber.

4 The exact order-four cutoff residual

Let $\theta = t\,d/dt$ and set

$$\mathcal{L} = \theta^4 - t\left(\theta + \frac{1}{3}\right)^2\left(\theta + \frac{2}{3}\right)^2.$$

Then $\mathcal{L}F = 0$.

Theorem 4.1 (Exact cutoff residual)**.** *For every $N \geq 1$,*

$$\boxed{\mathcal{L}P_N(t) = -N^4 c_N t^N.}$$

In particular, for $N = p^s$ the residual is divisible by p^{4s} in every coefficient ring in which $c_N t^N$ is integral.

Proof. The coefficient recurrence for F is

$$n^4 c_n = \left(n - \frac{2}{3}\right)^2 \left(n - \frac{1}{3}\right)^2 c_{n-1} \qquad (n \geq 1).$$

Applying $\mathcal{L}$ to P_N makes the coefficient of t^n vanish for $0 \leq n < N$. The only uncancelled term is produced by the shifted part at $n = N$, and equals $-N^4 c_N t^N$. □

For $N = p^s$, the differential residual has four powers of p before any comparison with a Frobenius solution. Estimating the error of such a comparison on the special fiber is a separate problem.

5 An ordinary unit-root criterion

In the rank-four Hodge structure at z_* the decomposition has the form

$$V_{z_*,\mathrm{B}} = \Lambda \oplus \Lambda^\perp, \qquad \Lambda_{\mathbb{C}} = V^{3,0} \oplus V^{0,3}, \qquad (\Lambda^\perp)_{\mathbb{C}} = V^{2,1} \oplus V^{1,2}.$$

Here B denotes the rational Betti realization, and $V^{i,j}$ denotes its Hodge components. We first record what the unit-root calculation gives under an explicit integral hypothesis.

Definition 5.1 (Arithmetic splitting package)**.** Fix an ordinary good prime $p \geq 5$ with $p \nmid a_p$. Assume the following data.

(A1) an integral rank-two Frobenius-stable direct summand W_p of the rank-four crystalline module at t_*, containing the holomorphic period direction;

(A2) Hodge weights $\{0, 3\}$ on W_p;

(A3) characteristic polynomial

$$\det(1 - T\varphi \mid W_p) = 1 - a_p T + p^3 T^2;$$

(A4) an integral comparison of the truncated jet $Y_p = (P_p, P_p', P_p'', P_p''')^{\mathsf{t}}$ at t_* with the Frobenius unit eigenvalue α_p on W_p, with first-coordinate error $P_p(t_*) - \alpha_p \in p^3\mathbb{Z}_p$.

Theorem 5.2 (Conditional splitting-to-supercongruence bridge)**.** *Let $p \geq 5$ and $p \nmid a_p$. If the hypotheses of Definition 5.1 hold, then*

$$P_p(t_*) \equiv a_p \pmod{p^3}.$$

Proof. Modulo p the polynomial $X^2 - a_p X + p^3$ has the simple nonzero root $a_p \bmod p$. Hensel's lemma gives its unit root α_p, and $a_p - \alpha_p = p^3/\alpha_p \in p^3\mathbb{Z}_p$. Adding the first-coordinate estimate in (A4) gives the assertion. □

Remark 5.3. The estimate in (A4) is an independent integral hypothesis. The rational splitting and knowledge of its characteristic polynomial identify the unit root at ordinary primes. They supply no estimate of the truncation error. If $p \mid a_p$, the reduction of the polynomial is X^2 and there is no unit root. Part IV uses Cayley–Hamilton on the holomorphic vector, so its result includes this case.

6 The exact arithmetic target

The decomposition of the rank-four constituent identified in Part IV, Theorem C.1, is

$$V^{\mathrm{ss}}_{z_*,\ell} \simeq \rho_{54.4.a.c,\ell} \oplus \rho_{54.2.a.a,\ell}(-1). \tag{1}$$

Equivalently, at every good prime,

$$\boxed{P_{3,p}(T) = (1 - a_pT + p^3T^2)(1 - pb_pT + p^3T^2).} \tag{2}$$

Here $P_{3,p}$ is the Euler polynomial of $V_{z_*,\ell}$. Period computations in [10] identified the candidate modular forms. A rational Hodge splitting alone does not establish the Galois action on those summands.

Two approaches to the semisimplified representation are relevant. One can construct a projector over $\mathbb{Q}$ and identify the representations on its image and complement. Alternatively, a proof of (2) at all good primes would identify the semisimplification by Chebotarev and Brauer–Nesbitt. The latter comparison alone does not locate the holomorphic vector in a specified de Rham summand. The tetrahedral projector in Part IV does locate this vector.

7 Residual monodromy at every auxiliary prime

The hypergeometric local system has standard companion matrices associated to

$$f(x) = (x^2 + x + 1)^2, \qquad g(x) = (x - 1)^4.$$

We use

$$A = \begin{pmatrix} 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & -2 \\ 0 & 1 & 0 & -3 \\ 0 & 0 & 1 & -2 \end{pmatrix}, \qquad B = \begin{pmatrix} 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & 4 \\ 0 & 1 & 0 & -6 \\ 0 & 0 & 1 & 4 \end{pmatrix}.$$

Put $e_4^T = (0, 0, 0, 1)$ and $v = (0, -2, 1, -2)^T$. Then

$$A - B = 3ve_4^T.$$

Theorem 7.1 (Complete residual-monodromy classification)**.** *Let ℓ be a prime and let $V = \mathbb{F}_\ell^4$.*

(a) If $\ell \neq 3$, the group generated by the reductions of A and B acts irreducibly on V.

(b) If $\ell = 3$, then $A \equiv B$, the common matrix is regular unipotent with characteristic and minimal polynomial $(x - 1)^4$, and its invariant subspaces are exactly the unique flag

$$0 \subset \ker(A - I) \subset \ker(A - I)^2 \subset \ker(A - I)^3 \subset V.$$

In particular there is no residual direct-sum decomposition $V = V_2 \oplus V_2'$ preserved by the geometric monodromy.

Proof. Consider the observability matrix

$$\mathcal{O} = \begin{pmatrix} e_4^T \\ e_4^T A \\ e_4^T A^2 \\ e_4^T A^3 \end{pmatrix} = \begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & -2 \\ 0 & 1 & -2 & 1 \\ 1 & -2 & 1 & 2 \end{pmatrix}, \qquad \det \mathcal{O} = 1.$$

Also consider the cyclic matrix

$$\mathcal{C} = (v, Av, A^2v, A^3v), \qquad \det \mathcal{C} = 81.$$

Let $W \subset V$ be invariant under both A and B, and assume $\ell \neq 3$. If e_4^T vanishes on W, then it vanishes on A^iW for $0 \leq i \leq 3$; invertibility of $\mathcal{O}$ gives $W = 0$. Otherwise choose $w \in W$ with $e_4^T w \neq 0$. Since $(A - B)w = 3(e_4^T w)v$, one has $v \in W$. The determinant 81 is nonzero modulo ℓ, so v, Av, A^2v, A^3v span V and $W = V$.

For $\ell = 3$, one has $A = B$ and A is the companion matrix of $(x-1)^4$. Hence V is the cyclic $\mathbb{F}_3[x]$-module $\mathbb{F}_3[x]/(x-1)^4$. Its submodules are the ideals $(x-1)^j$, which form the displayed unique flag. □

Corollary 7.2 (Uniqueness of the residual route)**.** *Suppose an integral arithmetic hypergeometric realization has geometric mod-3 monodromy generated by the reductions of A and B. Since geometric monodromy is normal in arithmetic monodromy, the arithmetic image normalizes it. The unique flag in Theorem 7.1(b) is characteristic and is therefore preserved by the arithmetic image.*

For $\ell \neq 3$, the standard companion lattice admits no rank-two sub-local system after reduction. For $\ell = 3$ its geometric invariant subspaces form the stated flag.

This calculation completes the auxiliary-prime search on the specified lattice. Lifting the flag to characteristic zero would require additional information about extension classes. Specializing the family can introduce new endomorphisms, as occurs for the tetrahedral symmetries at z_*.

8 The fiber-product and isogeny routes

The Hadamard-square presentation suggests a fiber-product model built from the cubic elliptic family. This makes low-degree fiberwise isogenies tempting. The simplest versions fail.

8.1 The Fricke-3 mismatch

In the standard Hesse coordinate α, the Fricke involution is

$$\alpha \longmapsto 1 - \alpha,$$

whereas the involution induced by the proposed attractor fiber product is

$$\alpha \longmapsto -\frac{1}{8\alpha}.$$

These are not the same rational map. Thus the splitting is not explained by the naive identification of the two elliptic factors through the standard Fricke 3-isogeny.

8.2 An exact degree-2 obstruction

The Hesse j-function in this normalization is

$$j(\alpha) = \frac{27(1 + 8\alpha)^3}{\alpha(1 - \alpha)^3}.$$

Let $\Phi_2(X, Y)$ denote the classical level-2 modular polynomial.

Proposition 8.1 (No generic degree-2 fiberwise isogeny)**.** *The rational function*

$$\Phi_2\Big(j(\alpha), j\Big(-\frac{1}{8\alpha}\Big)\Big)$$

is nonzero. More precisely, its numerator is

$$\begin{aligned}&-19683(8\alpha^2+20\alpha-1)^4(64\alpha^4-112\alpha^3+60\alpha^2+14\alpha+1)\\&\qquad\times Q_{12}(\alpha),\end{aligned}$$

where

$$\begin{aligned}Q_{12}(\alpha) = {}& 262144\alpha^{12}+189726720\alpha^{11}+69710610432\alpha^{10}-232501817344\alpha^9\\&+254561642496\alpha^8-48116259072\alpha^7-43846988736\alpha^6+6014532384\alpha^5\\&+3977525664\alpha^4+454105112\alpha^3+17019192\alpha^2-5790\alpha+1.\end{aligned}$$

The denominator is $\alpha^3(\alpha-1)^9(8\alpha+1)^9$.

Proof. Substitute the displayed j-functions into the classical polynomial

$$\begin{aligned}\Phi_2(X,Y) = {}& X^3+Y^3-X^2Y^2+1488XY(X+Y)-162000(X^2+Y^2)\\&+40773375XY+8748000000(X+Y)-157464000000000\end{aligned}$$

and factor over $\mathbb{Z}[\alpha]$. Appendix A reproduces the exact computation. □

The obstruction concerns a generic degree-2 isogeny of the two elliptic fibers. The correspondence in Part IV acts simultaneously on the pencils and their common base.

9 Residual hypotheses in a Frobenius comparison

At a fixed fiber, a finite set of Frobenius traces can suffice after the determinant, the residual representation, and its possible deviations have been determined. A mod-3 flag for the generic monodromy does not give those data at z_*. A naive Livné test based only on quadratic characters unramified outside $2, 3$ therefore leaves an implication to prove. Theorem C.1 in Part IV instead compares the already constructed rank-two representations at the auxiliary prime 2. Its residual image and the possible deviation fields specify the finite set of test primes.

10 A criterion using a Hodge projector

Let $e_{\mathrm{H}} \in \mathrm{End}_{\mathrm{HS}}(V_{z_*,\mathrm{B}})$ denote the projector onto $V^{3,0} \oplus V^{0,3}$ along its orthogonal complement. Extending it by zero on the orthogonal complement of $V_{z_*,\mathrm{B}}$ defines a Hodge class in $H^6(X_{z_*} \times X_{z_*}, \mathbb{Q})(3)$.

Proposition 10.1 (Compatible realizations of a projector)**.** *Suppose e_{H} is realized by a motivated projector defined over $\mathbb{Q}$, or by a descended absolute-Hodge projector with compatible Galois and crystalline realizations. Then its rank-two image is stable in the ℓ-adic realizations and under Frobenius in the good crystalline fibers. At an ordinary good prime, if these summands have the stated modular polynomial and the integral hypotheses of Definition 5.1 hold, then $P_p(t_*) \equiv a_p \pmod{p^3}$.*

Proof. Realization preserves the idempotent equation and its rank. Over $\mathbb{Q}$, the realization of the projector commutes with the Galois action and, at good primes, with Frobenius. The final assertion uses the integral comparison in (A4) and Theorem 5.2. □

This criterion records an alternative to finding a correspondence explicitly. Appendix A gives an algebraic correspondence on the parabolic constituent.

11 The finite-field hypergeometric route

A direct proof of

$$P_{3,p}(T) = (1 - a_pT + p^3T^2)(1 - pb_pT + p^3T^2)$$

would bypass the global projector. Hypergeometric motives admit trace descriptions through Gauss and Jacobi sums, and the special value $-1/8$ suggests cubic and quadratic transformations. The search found finite-field Clausen and Kummer identities near this value, but not an identity that factors the entire degree-four Euler polynomial into the two specified modular quadratics.

The exact statement needed is stronger than a single trace identity. One must control enough power traces (or trace and determinant data on a rank-two summand) to identify the whole quartic factor at every good prime.

12 Exact numerical checks

The verification program in Appendix A computes the truncation in exact rational arithmetic and reduces it modulo p^3. The first residues are

p	5	7	11	13	17	19	23	29	31
$P_p(-1/8) \bmod p^3$	3	29	-57	20	-72	-106	174	-210	47.

For $p = 5, 7$, these equal the displayed coefficients $a_5 = 3$ and $a_7 = 29$ of 54.4.*a.c*. The remaining entries are recorded as exact project checks; their identification with the corresponding LMFDB coefficients is not used as a proof.

13 Development of the arithmetic argument

(i) The formal descent problem suggested isolating the special fiber as the source of the extra power of p.

(ii) The Picard–Fuchs recurrence gave the exact residual $-N^4c_Nt^N$. The ordinary rank-two polynomial separated comparison with the truncated period from comparison with the Fourier coefficient.

(iii) Numerical period and Frobenius data suggested the weight-four and twisted weight-two factors of (2).

(iv) The Fricke involution and the degree-two modular polynomial eliminated two simple fiberwise explanations. The companion matrices then gave the generic residual classification at every auxiliary prime.

(v) The simultaneous transformations of the Hesse pencils at z_* produced a rational projector. The four-prime comparison on the rank-two summands identified their modular factors. Interpolation of the shifted factorials in Part IV completed the integral comparison for every good prime.

14 From the projector to the truncation

The two results in Part IV can be used separately. Its Theorem C.1 gives (1) and (2). Its Lemma 4.1 identifies the first entry of $p^3\Phi^{-1}$ on the rational lift with $P_p(t_*)$ modulo p^3, and controls the integral lattice in which the characteristic polynomial acts. Together they give Theorem 1.1. This route applies to ordinary and nonordinary good primes alike.

A Exact verifier

The following program verifies the coefficient recurrence, the first truncation residues, the determinants used in Theorem 7.1, and the exact modular-polynomial factorization of Proposition 8.1. It uses exact arithmetic throughout.

```
#!/usr/bin/env python3
"""Exact checks for the AESZ4 level-54 attractor dossier.

Standard-library checks are used where possible; SymPy is used only for exact
symbolic polynomial factorization and matrix determinants.
"""
from __future__ import annotations

import json
from fractions import Fraction
from math import factorial
from pathlib import Path

import sympy as sp

OUT = Path(__file__).with_name("verification_output.json")


def generalized_binomial(a: Fraction, k: int) -> Fraction:
    result = Fraction(1)
    for j in range(k):
        result *= a - j
    return result / factorial(k)


def truncated_sum_mod(p: int) -> int:
    modulus = p**3
    total = Fraction(0)
    for k in range(p):
        total += (
            generalized_binomial(Fraction(-1, 3), k) ** 2
            * generalized_binomial(Fraction(-2, 3), k) ** 2
            * Fraction(-1, 8) ** k
        )
    return (total.numerator * pow(total.denominator, -1, modulus)) % modulus


def picard_fuchs_check(max_n: int = 12) -> bool:
    # c_n=(1/3)_n^2(2/3)_n^2/(n!)^4 and
    # n^4 c_n=(n-2/3)^2(n-1/3)^2 c_{n-1}.
    c = [Fraction(1)]
    for n in range(1, max_n + 1):
        c.append(
            c[-1]
            * Fraction(3 * n - 2, 3) ** 2
            * Fraction(3 * n - 1, 3) ** 2
            / n**4
        )
    return all(
        n**4 * c[n]
```

```
        == Fraction(3 * n - 2, 3) ** 2
        * Fraction(3 * n - 1, 3) ** 2
        * c[n - 1]
        for n in range(1, max_n + 1)
    )


def residual_monodromy_data() -> dict[str, object]:
    A = sp.Matrix(
        [
            [0, 0, 0, -1],
            [1, 0, 0, -2],
            [0, 1, 0, -3],
            [0, 0, 1, -2],
        ]
    )
    B = sp.Matrix(
        [
            [0, 0, 0, -1],
            [1, 0, 0, 4],
            [0, 1, 0, -6],
            [0, 0, 1, 4],
        ]
    )
    e4 = sp.Matrix([[0, 0, 0, 1]])
    v = sp.Matrix([0, -2, 1, -2])
    observability = sp.Matrix.vstack(*[e4 * (A**i) for i in range(4)])
    cyclic = sp.Matrix.hstack(*[(A**i) * v for i in range(4)])
    assert A - B == 3 * v * e4
    assert observability.det() == 1
    assert cyclic.det() == 81
    x = sp.symbols("x")
    assert sp.factor(A.charpoly(x).as_expr()) == (x**2 + x + 1) ** 2
    assert sp.factor(B.charpoly(x).as_expr()) == (x - 1) ** 4
    return {
        "A_minus_B": [[int(z) for z in row] for row in (A - B).tolist()],
        "observability_det": int(observability.det()),
        "cyclic_det": int(cyclic.det()),
        "charpoly_A": str(sp.factor(A.charpoly(x).as_expr())),
        "charpoly_B": str(sp.factor(B.charpoly(x).as_expr())),
        "mod3_flag_dimensions": [1, 2, 3, 4],
    }


def phi2_factorization() -> dict[str, str]:
    a, X, Y = sp.symbols("a X Y")
    phi2 = (
        X**3
        + Y**3
        - X**2 * Y**2
        + 1488 * X * Y * (X + Y)
        - 162000 * (X**2 + Y**2)
        + 40773375 * X * Y
        + 8748000000 * (X + Y)
        - 157464000000000
    )
    j = lambda z: 27 * (1 + 8 * z) ** 3 / (z * (1 - z) ** 3)
    expression = sp.factor(
        sp.together(phi2.subs({X: j(a), Y: j(-sp.Rational(1, 8) / a)}))
    )
    numerator, denominator = sp.fraction(expression)
    assert numerator != 0
    return {
        "numerator": str(sp.factor(numerator)),
        "denominator": str(sp.factor(denominator)),
    }
```

```
def main() -> None:
    residues = {str(p): truncated_sum_mod(p) for p in [5, 7, 11, 13, 17, 19, 23, 29, 31]}
    expected_signed = {
        "5": 3,
        "7": 29,
        "11": -57,
        "13": 20,
        "17": -72,
        "19": -106,
        "23": 174,
        "29": -210,
        "31": 47,
    }
    for p_string, signed in expected_signed.items():
        p = int(p_string)
        assert residues[p_string] == signed % (p**3)

    report = {
        "picard_fuchs_coefficient_recurrence": picard_fuchs_check(),
        "truncated_sum_mod_p3": residues,
        "signed_representatives": expected_signed,
        "residual_monodromy": residual_monodromy_data(),
        "phi2_substitution": phi2_factorization(),
    }
    OUT.write_text(json.dumps(report, indent=2) + "\n", encoding="utf-8")
    print(json.dumps(report, indent=2))


if __name__ == "__main__":
    main()
```

B Verification output

```
Picard--Fuchs recurrence: PASS
observability determinant: 1
cyclic determinant: 81
characteristic polynomial of A: (x^2+x+1)^2
characteristic polynomial of B: (x-1)^4
mod-3 invariant flag dimensions: 1,2,3,4
truncated residues (signed):
p=5: 3; p=7: 29; p=11: -57; p=13: 20; p=17: -72;
p=19: -106; p=23: 174; p=29: -210; p=31: 47.
Phi_2 substitution numerator: nonzero; exact factors given above.
```

C Source and verification discipline

The verifier checks the cutoff recurrence, the companion matrices, the sample truncations, and the modular-polynomial calculation. The arithmetic decomposition uses the geometric projector and the Galois comparison in Part IV. The verification range for the good-prime congruence is recorded separately in that part.

PART IV

The Level-54 Attractor Supercongruence for Every Good Prime

12 November 2026

Abstract

We obtain the congruence of Beukers at the level-54 attractor for every prime $p \geq 5$, with no ordinarity hypothesis. Interpolating complete blocks in the shifted factorial period gives an overconvergent inverse Frobenius on the cyclic Picard–Fuchs frame. Its first coordinate is the truncated period modulo p^3. Its four rows have divisibilities $1, p, p^2, p^3$, and its determinant at a fixed fiber is p^6. These facts turn the characteristic polynomial of the rank-two component containing the holomorphic form into the required truncation congruence. We include the construction and modular identification of that component and an independent finite calculation. The prime 2 in the printed formulation fails by direct evaluation.

1 The good-prime congruence

Write $A_n = (3n)!/(n!)^3$, $c = -1/5832$, and $S_p = \sum_{n=0}^{p-1} A_n^2 c^n$. Let a_p be the coefficient of q^p in the normalized newform f labeled 54.4.a.c. It has weight four, level 54, trivial character, and begins $q + 2q^2 + 4q^4 + 3q^5 + \cdots$.

Theorem 1.1. *For every prime $p \geq 5$,*

$$\sum_{k=0}^{p-1} \binom{-1/3}{k}^2 \binom{-2/3}{k}^2 \left(-\tfrac{1}{8}\right)^k = S_p \equiv a_p \pmod{p^3}.$$

The equality follows from $(1/3)_k(2/3)_k/(k!)^2 = A_k/27^k$. The theorem treats the good primes of Conjecture 1.28 of Beukers [7]. Its printed quantifier $p \neq 3$ includes 2, where $S_2 = 161/162$, $a_2 = 2$, and $v_2(S_2 - a_2) = -1$.

Let V_z be the four-dimensional summand of Bönisch's Hesse fiber-product family [18, §5.1]. Its ambient threefold has 40-dimensional H^3. At $z = c$, Appendix A constructs a rational rank-two projector whose image W contains the holomorphic form Ω_c. Appendix C identifies the trace and determinant of its Frobenius as a_p, p^3. We use the convention that twisting by (-1) multiplies good-prime eigenvalues by p.

The passage from W to the truncation occupies Sections 2–4. The matrix is built over the whole smooth parameter locus; only its final Cayley–Hamilton identity is imposed on the special fiber.

2 A Frobenius operator from the shifted coefficients

2.1 Interpolation of the complete factorial blocks

Write $\theta = z\, d/dz$ and use the rising factorial $(x)_n = \prod_{i=0}^{n-1}(x+i)$. The differential equation in the factorial coordinate is

$$\theta^4 y = 729z(\theta + \tfrac{1}{3})^2(\theta + \tfrac{2}{3})^2 y. \tag{1}$$

It is nonsingular on the base $\mathbf{P}^1 - \{0, 1/729, \infty\}$. All local estimates that follow include $p = 5$. Fix $p \geq 5$ and put

$$P(T) = \prod_{\nu=1}^{p-1} (1 + pT/\nu).$$

Let $H(T)$ be the discrete primitive of $\log P(T)$, normalized by $H(0) = 0$. To construct it explicitly, write

$$\log P(T) = \sum_{m \geq 1} d_m T^m, \quad d_m = (-1)^{m+1} \frac{p^m}{m} \sum_{\nu=1}^{p-1} \nu^{-m}.$$

Replace each T^m by $\sum_{j=0}^{m} \left\{ {m \atop j} \right\} T^{\underline{j+1}}/(j+1)$, where the braces are Stirling numbers of the second kind and the underline is the falling factorial. Translation by one minus the identity sends this polynomial to T^m. For $m = 1, 2$ the d_m belong to $p^3\mathbf{Z}_p$ by Wolstenholme and the power sum in $\mathbf{F}_p$. For $m \geq 3$ every coefficient of the replacement times d_m has valuation at least $m - v_p(m) - \lfloor \log_p(m+1) \rfloor \geq 3$. The lower bound grows linearly apart from logarithmic corrections and the degree is at most $m + 1$. Thus $H \in p^3\mathbf{Z}_p\langle T \rangle$ overconverges and has sufficiently small norm for exponentiation on a disk strictly larger than the unit disk. More explicitly, for degree $m + 1$ each contribution to H has valuation at least $m - v_p(m) - \lfloor \log_p(m+1) \rfloor$, and the low degrees have valuation at least three. Choosing a sufficiently small $\epsilon > 0$ makes the Gauss norm of H on $|T| \leq p^\epsilon$ strictly smaller than $p^{-1/(p-1)}$. This proves norm convergence of the exponential in the corresponding Tate algebra. Define

$$b_0(T) = \exp(2[H(3T) - 3H(T)]), \qquad b_k(T) = b_0(T) \left(\frac{(1+3pT)_{3k}}{(1+pT)_k^3} \right)^2 \quad (1 \leq k < p).$$

In particular $b_0(0) = 1$, $b_k(0) = ((3k)!/(k!)^3)^2$ and $[T^d]b_k(T) \in p^{\min(d,3)}\mathbf{Z}_p$ for all $d \geq 0$. In this estimate the numerator is left unnormalized even where one of its constants $1, \ldots, 3k$ is divisible by p. All denominators have unit constant. The factors are overconvergent on a common strict neighborhood of the unit disk for any given p.

For $a_j(u) = ((1+3u)_{3j}/(1+u)_j^3)^2$, splitting off factors at multiples of p gives the identity of germs at $u = 0$

$$a_{pj+k}(pu) = a_j(u) \frac{b_k(j+u)}{b_0(u)}.$$

Indeed the products of the nonmultiples of p in the complete blocks are products of $P(h+3u)$ and $P(h+u)$, respectively. The constant $(p-1)!$ of each block cancels between numerator and denominator, and summing the logarithms yields the quotient $b_0(j+u)/b_0(u)$. The remaining $3k$ and k factors yield $b_k(j+u)/b_0(j+u)$. This proof works even for the indices j with p-adic denominators in their individual u-Taylor coefficients, since $a_j(u)$ has been canceled before the block interpolation.

2.2 The operator on the cyclic frame

Use the cyclic vector $\omega = \Omega_z$ of the Picard–Fuchs equation, let $\theta = z\,d/dz$, and $e_j = \theta^j\omega$, $0 \leq j \leq 3$. Define an operator from the module at z into its pullback at z^p by putting $w = z^p$ and setting

$$C(e_j(z)) = \sum_{k=0}^{p-1} z^k b_k(\theta_w)(k + p\theta_w)^j \omega(w) \Big|_{w=z^p} \qquad (0 \leq j \leq 3). \tag{2}$$

Here θ inside the differential operator acts at the target argument before substitution. The differential equation eliminates $\theta^4\omega$ with coefficients in $\mathbf{Z}_p[z,(1-729z)^{-1}]$. Each additional differentiation in this frame increases pole order and numerator degree by at most a fixed constant, since the derivative of a rational entry is added to its product with the companion matrix of (1). Hence the preceding overconvergent series for each column converges over the weak completion, by taking a sufficiently small strict neighborhood of the good locus. The reductions do not lose p-adic integrality. A column written in the ordered target frame has its entries in $\mathbf{Z}_p, p\mathbf{Z}_p, p^2\mathbf{Z}_p, p^3\mathbf{Z}_p$, respectively, at any good integral point. To see this directly, the coefficient of θ_w^h in $b_k(\theta_w)(k+p\theta_w)^j$ has valuation at least $\min(h,3)$. For $h=0,1,2$ the indicated monomial is already a frame element, and for $h\geq 3$ all terms after reduction are divisible by p^3. In particular row i of the matrix, where the target frame element is $e_i(w)$, is divisible by p^i. At precision p^3 its entry in row zero and column zero is $\sum_{k=0}^{p-1}((3k)!/(k!)^3)^2z^k$.

Write $G(z,u)=z^u\sum_{n\geq 0}a_n(u)z^n$ modulo u^4, using formal logarithms. Its four u-coefficients are a fundamental system of solutions of (1): the indicial term is u^4z^u and the recurrence at each positive index is exactly the recurrence for $a_n(u)$. Their leading logarithmic terms $1, \log z, (\log z)^2/2$ and $(\log z)^3/6$ are linearly independent. Evaluating the target column against these four solution jets is exactly the coefficient identity

$$\sum_{k=0}^{p-1} z^k[b_k(\theta)(k+p\theta)^jG(\,\cdot\,,u)](z^p) = b_0(u)\,\theta^jG(z,pu).$$

All computations in this identity are coefficientwise at $z=0$ with coefficients in $(\mathbf{Q}_p[u]/(u^4))[\log z]$. Running over the four solution jets gives an identity of fundamental matrices. The matrix acting on the jets on the right is constant in z, so differentiation proves horizontality, including the relation at derivative order four. Its determinant is p^6, since $u\mapsto pu$ acts by $1,p,p^2,p^3$ and multiplication by $b_0(u)$ has unit diagonal. The indicial initial conditions normalize the θ-Wronskian to one at $z=0$. Abel's identity applied to (1) then gives the Wronskian $(1-729z)^{-2}$. This gives $\det C = p^6(1-729z^p)^2/(1-729z)^2$ away from the deleted disks. In particular after Taylor transport at a rational fiber the determinant is p^6.

3 Normalizing the geometric Frobenius

Write $\theta = z\,d/dz$ and $\lambda = 729$. The algebraic connection of the Hesse family on $B=\mathbf{P}^1\setminus\{0,\lambda^{-1},\infty\}$ is the rank-four hypergeometric connection used above. In particular the vector e_0 in its cyclic frame is the normalized holomorphic three-form Ω_z. This normalization matters for the coefficient calculation. Bönisch [18, §5.1] constructs Ω_z by residues of the two cubic pencils, computes its period at 0 as $\sum A_n^2z^n$, and identifies $\Omega_z, \theta\Omega_z, \theta^2\Omega_z, \theta^3\Omega_z$ as a cyclic frame. Its companion equation has no additional singular points.

Fix $p\geq 5$. The Hesse surfaces and their projective fiber product give a smooth projective family of threefolds over the smooth parameter locus after reduction. To see good reduction in families, one can first adjoin ζ_3 and $z^{1/3}$ on this locus. The base points of each Hesse pencil are nine distinct points and its four singular fibers are triangles. The singular fibers of the two pencils are disjoint precisely away from $\lambda z=1$ (and the degenerate parameters $0,\infty$). Thus the total threefold in each fiber is smooth. The vertical restriction kernels and the cyclic average defining V are functorial in these families and descend. In particular relative cohomology supplies V with a geometric Frobenius $\Phi:\sigma^*V\to V$ for $\sigma(z)=z^p$. For clarity, the smooth projective overconvergence theorem applies to the ambient Gauss–Manin connection [21, Corollary 6.4]. On a finite étale parameter cover splitting the components, the restriction maps to the ruled components commute

with this Frobenius. Taking their kernel and cyclic invariants inside the category of overconvergent isocrystals, followed by descent, gives the same overconvergence for V. At a rational lift in a residue disk, Taylor transport identifies Φ with the Frobenius on the crystalline cohomology of that fiber in the de Rham realization.

Lemma 3.1. *The horizontal map $C : V \longrightarrow \sigma^{*}V$ constructed from the coefficient blocks equals $p^{3}\Phi^{-1}$.*

Proof. The inverse of C is a Frobenius intertwiner on a strict neighborhood, by the determinant formula and the construction above. Kedlaya's uniqueness theorem for hypergeometric intertwiners [20, Lemma 2.3.6 and Remark 4.1.3] applies. In the coordinate λz the upper parameters are $1/3, 1/3, 2/3, 2/3$ and the lower parameters are $1, 1, 1, 1$. No upper-minus-lower difference is integral. The singularities are distinct modulo p, the exponents are p-integral, and multiplication by p permutes the upper classes and fixes the lower classes modulo $\mathbf{Z}$. The coordinate rescaling changes the Frobenius lift by the unit $\lambda^{1-p} \equiv 1 \bmod p$, which is allowed by the Taylor change-of-lift formula [20, Remark 2.3.7]. Hence $C = \kappa\, p^{3}\Phi^{-1}$ for some $\kappa \in \mathbf{Q}_{p}^{\times}$.

At any smooth rational Teichmüller parameter $\widehat{z}$, where $\widehat{z}^{\,p} = \widehat{z}$, the determinant of C is p^{6}. The alternating perfect pairing of the polarized rank-four V has similitude p^{3} so $\det \Phi = p^{6}$. Thus $\kappa^{4} = 1$. To determine which root it is, let

$$P(z) = \sum_{0 \le k \le \lfloor (p-1)/3 \rfloor} A_{k}^{2} z^{k} \quad \text{in } \mathbf{F}_{p}[z].$$

At a Teichmüller parameter the row estimates of the explicit matrix give $\operatorname{Tr} C \equiv P(z) \pmod p$. The same holds for $\operatorname{Tr} \Phi$. Here is the point-count justification over all smooth parameters. In the first Hesse pencil use the three coordinates proportional to (X^{3}, Y^{3}, Z^{3}) and summing to one; in the second use the coordinates proportional to (X^{3}, Y^{3}, zZ^{3}) and summing to one. Their respective products are u and z/u. The degree-three quotient in each elliptic fiber and the same cyclic descent as in Appendix B identify the rank-four parabolic term throughout this locus. This description is defined over the base before choosing a cube root of z. The product-moment count there, with z in place of c, gives $\operatorname{Tr} \Phi \equiv P(z)$. Self-duality pairs every eigenvalue α with p^{3}/α, so $\operatorname{Tr}(p^{3}\Phi^{-1}) = \operatorname{Tr} \Phi$. Finally $P(0) = 1$ and $\deg P \le (p-1)/3 < p-2$; at least one of the $p-2$ smooth nonzero residues has $P(z) \neq 0$. Reduction there gives $\kappa \equiv 1 \bmod p$. As $p \nmid 4$, this forces $\kappa = 1$. □

4 The rational lift and the third digit

We require the value at the characteristic-zero parameter c, whose de Rham holomorphic vector lies in the geometric rank-two space W. We therefore compare the fibers at c^{p} and c. Put $L = \log(c^{p-1}) \in p\mathbf{Z}_{p}$. In the cyclic coordinates at c, Taylor transport from c^{p} to c sends the vector $\theta^{j}\Omega$ evaluated at c^{p} to

$$\sum_{h \ge 0} \frac{L^{h}}{h!}\, (\theta^{h+j}\Omega)(c).$$

This follows by pairing against a horizontal section in the residue disk and expanding its scalar functions at c in the coordinate $\log(z/c)$. The same formula follows directly by Taylor transport for the Gauss–Manin connection. The companion relation reduces every $\theta^{n}\Omega$ integrally in the cyclic lattice at c, since $1 - \lambda c = 9/8$ is a p-adic unit. The series converges, and $v_{p}(L^{h}/h!) \ge \min(h, 3)$ for $p \ge 5$. Denote the transported endomorphism of V_{c} by C_{c}.

Lemma 4.1. *In the ordered basis (e_0, e_1, e_2, e_3) at c, with $e_i = \theta^i\Omega$, put $\Lambda = \sum_i \mathbf{Z}_p e_i$. Then*

$$C_c = \operatorname{diag}(1, p, p^2, p^3)\, U, \qquad U \in \mathrm{GL}_4(\mathbf{Z}_p), \qquad (C_c)_{00} \equiv S_p \pmod{p^3}.$$

Proof. Before transport, the coefficient of $\theta^j\Omega$ in the unreduced series of each column is divisible by $p^{\min(j,3)}$. Multiplication by the Taylor series preserves this estimate: at total order $j + h$ the valuations add to at least $\min(j + h, 3)$. All terms of order three or more are integral multiples of p^3 after reduction by the companion equation. Orders zero, one and two are already basis vectors. This proves the four row divisibilities. For the first column the order-zero coefficient is exactly $\sum_{k<p} c^k b_k(0) = S_p$. Lower-order Taylor corrections involve e_1 and e_2 only. Thus reduction of the first coordinate changes S_p by a multiple of p^3. By Lemma 3.1, C_c is the geometric $p^3\Phi_c^{-1}$, and its determinant is p^6 at this fiber as well. Dividing row i by p^i gives an integral matrix of determinant one. □

Proof of Theorem 1.1. The averaging correspondence in Appendix A is defined over $\mathbf{Q}$ on V_c and has good reduction for $p \geq 5$. Functorial crystalline comparison at the lift c identifies its de Rham image with a Frobenius-stable rank-two space. Its holomorphic line contains e_0. The comparison in Appendix C gives the characteristic polynomial of Φ_c on this space as $X^2 - a_p X + p^3$. The same polynomial holds for $C_c = p^3\Phi_c^{-1}$, since the two roots have product p^3. Cayley–Hamilton applied on this space gives the vector identity

$$C_c e_0 + p^3 C_c^{-1} e_0 = a_p e_0.$$

By Lemma 4.1, $C_c^{-1} e_0 = U^{-1} e_0 \in \Lambda$. It follows that $C_c e_0 \equiv a_p e_0 \bmod p^3\Lambda$. Taking its first coordinate and using the same lemma proves the congruence. This argument works for all good primes, independently of the slopes of this rank-two factor: the integrality used here is in the explicit cyclic-coordinate lattice. □

Corollary 4.2. *Put $m = \lfloor (p-1)/3 \rfloor$ and $q_p(c) = (c^{p-1} - 1)/p$. With $H_j = \sum_{i=1}^{j} 1/i$ and $H_0 = 0$, every good prime satisfies*

$$\sum_{k=0}^{m} A_k^2 c^k \bigl(q_p(c) + 6(H_{3k} - H_k)\bigr) \equiv 0 \pmod{p}.$$

Proof. Read row one of $C_c e_0 \equiv a_p e_0 \bmod p^3\Lambda$ at precision p^2. Its unreduced coefficient is the coefficient of T in $\exp(LT) \sum_{k<p} c^k b_k(T)$. Reducing higher powers of θ does not affect this coefficient modulo p^2. We have $L \equiv p q_p(c) \bmod p^2$ and $b_0'(0) = 0$. For $3k < p$, logarithmic differentiation of the finite factorial quotient gives $b_k'(0) = 6pA_k^2(H_{3k} - H_k)$. For $3k \geq p$ at least one squared factor $(ph + 3pT)^2$ occurs in the numerator and the entire series is divisible by p^2. Dividing the row-one congruence by p proves the claim without any division by S_p. □

5 Exact checks

The accompanying code tests the Frobenius matrix and the newform independently. The first tests the transported matrix at every smooth nonzero parameter in $\mathbf{F}_p$ for the 23 primes $5 \leq p \leq 97$ (1,009 fibers). It compares its trace modulo p^3 with an independent count of pairs of toric cubics. At $z = c$ it also tests the vanishing of its two off-diagonal first-column entries modulo p^3 in rows one and two; row three is divisible by p^3 by construction.

The second computes the Fourier coefficients of f without using the hypergeometric period. The 21 eta quotients used in `verify_finite.py` give a basis for $S_4(\Gamma_0(54))$; the checks of their

cusp orders, character and independence, and the Sturm bound 36, are made over $\mathbf{Q}$. Imposing $U_2 = 2$, $U_3 = 0$ and $T_5 = 3$ leaves a one-dimensional joint eigenspace, which is normalized by its leading coefficient. For the longer computation we expand twice this form modulo 2^{64} using Euler's pentagonal number theorem. The bound $|a_p| \le 2p^{3/2}$ makes the signed lift unique in the tested range. The ratio of successive truncation terms is $-(3k-1)^2(3k-2)^2/(648k^4)$; clearing prefix denominators gives a division-free check modulo p^3. Terms with $3k \ge 2p$ vanish at this precision. All 9,590 primes $5 \le p \le 100{,}000$ passed. For example, $a_{99991} = 49{,}718{,}099$. In the separate computation modulo p^4, all 667 primes $5 \le p \le 5000$ had $v_p(S_p - a_p) = 3$. Already $S_5 \equiv 128 \bmod 625$ and $a_5 = 3$ prove that the exponent three cannot be increased uniformly. Every prime up to 100,000 in this computation had $p \nmid a_p$; the proof of the theorem makes no assumption about the existence or distribution of nonordinary primes.

A The tetrahedral projector

A.1 The Hesse cover

We also write $z_* = c$. Bönisch [18, §5.1] constructs an elliptic surface S_z with equation $s_0XYZ = t_0(X^3 + Y^3 + zZ^3)$ and forms $S_1 \times_{\mathbf{P}^1} \epsilon^* S_z$, where ϵ interchanges s_0 and t_0. Set $s = s_0/(3t_0)$. Since $c = (-1/18)^3$, putting $Z_{\text{new}} = (-1/18)Z_{\text{old}}$ in the second cubic identifies this threefold over $\mathbf{Q}$ on the smooth locus with the product of

$$H_s : \; X^3 + Y^3 + Z^3 = 3sXYZ, \qquad H_{I(s)}, \qquad I(s) = -2/s.$$

Let $D = \{\infty, 1, \zeta, \zeta^2, 0, -2, -2\zeta, -2\zeta^2\}$, where ζ is a primitive third root of unity. On $j : \mathbf{P}^1 \setminus D \hookrightarrow \mathbf{P}^1$ write $L = R^1\pi_*\mathbf{Q}_\ell$ for the first Hesse local system, and set

$$U = H^1(\mathbf{P}^1_{\overline{\mathbf{Q}}}, j_*(L \otimes I^*L)). \tag{3}$$

This is the nonvertical component U_{z_*} in [18, §§4.3, 5.1]. Here are the details of that identification. In Leray for the threefold over $\mathbf{P}^1$ the term $E_2^{1,2}$ has no incoming or outgoing differentials. Fiberwise Künneth expresses R^2 as the two elliptic R^2 sheaves and $j_*(L \otimes I^*L)$, since at most one factor is singular over each point. Each elliptic R^2 maps onto the constant Tate line, with skyscraper kernel, and so has zero H^1. Furthermore $E_2^{2,1} = 0$: here $R^1 = j_*(L \oplus I^*L)$, and the dual elliptic systems have no global invariants. The Leray filtration therefore identifies $E_2^{1,2}$ with a subspace of H^3 annihilated by restriction to fibers. It lies in Bönisch's eight-dimensional kernel of the vertical component restrictions. Equality follows by the dimension calculation for parabolic cohomology below. The Euler characteristic gives $\dim U = 8$: the tensor has rank four, its invariant stalk at each puncture has dimension two, and its global invariants and dual invariants vanish. Indeed, the two irreducible elliptic local systems have different bad loci, hence are not geometrically isomorphic, even after dualizing. This also gives $H^0(\mathbf{P}^1, j_*(L \otimes I^*L)) = H^2(\mathbf{P}^1, j_*(L \otimes I^*L)) = 0$.

A.2 Two simultaneous transformations

Consider

$$d(s) = \zeta s, \qquad m(s) = \frac{s+2}{s-1}.$$

Then $Id = d^{-1}I$ and $Im = mI$. On projective triples let $D_0 = \operatorname{diag}(\zeta^2, 1, 1)$ and $F = (\zeta^{ij})_{0 \le i,j \le 2}$. Lifts on the pair of elliptic curves are $(D_0, \bar{D}_0)$ for d and $(F, \bar{F})$ for m. For the latter this follows by expanding the cube sum and product of the three entries of $F(X, Y, Z)^{\mathrm{t}}$: they are

$3(X^3+Y^3+Z^3+6XYZ)$ and $X^3+Y^3+Z^3-3XYZ$. Thus the new Hesse parameter is $m(s)$. These projectivities preserve the base locus of the pencil as a set, so also act on the resolved rational elliptic surfaces. On $\{\infty, 1, \zeta, \zeta^2\}$ the two base maps act as a three-cycle and a double transposition. They generate A_4.

There is a minor distinction between matrices on the pencil and their actions on the tensor. A word acting trivially on the base induces an automorphism of the generic elliptic curve, which is a translation followed by [1] or [−1], since its j-invariant is nonconstant. The sign on the second factor is the same: the second word is obtained from the first by conjugating ζ and evaluating at $I(s)$, and both signs are rational. Thus such a word acts trivially on $L \otimes I^*L$. We obtain an action of A_4 on U and on its Betti and de Rham counterparts. Conjugation sends d to d^{-1}. It also preserves the action of m: conjugation interchanges F and $\bar{F} = FP$, where P exchanges the last two coordinates. The involution P acts by -1 on H^1; it fixes a point and is nontrivial on the generic fiber. The two signs cancel on the tensor. Consequently the two projectors

$$e_C = \frac{1+d+d^2}{3}, \qquad e = \frac{1}{12}\sum_{h\in A_4} h \tag{4}$$

commute with $G_{\mathbf{Q}}$. The averaging takes place on cohomology by algebraic correspondences; $e_C e = e e_C = e$. Bönisch identifies $V = e_C U$ with V_{z_*}, of Hodge numbers $(1,1,1,1)$.

A.3 The two ranks and their Hodge types

Bönisch computes $\mathrm{Tr}(d \mid U) = \mathrm{Tr}(d^{-1} \mid U) = 2$. For m the only fixed base points are $1 \pm \sqrt{3}$, both smooth. At either point $F^2 = 3P$ in homogeneous coordinates, and hence the square of the fiber action on H^1 is -1. The action has determinant 1, since it is an elliptic automorphism of degree one. Its eigenvalues are $i, -i$, and its trace vanishes. The conjugate factor has trace zero as well. The two fixed points are simple. The Lefschetz formula for the equivariant constructible sheaf in (3) now yields $\mathrm{Tr}(m \mid U) = 0$, using the vanishing of H^0 and H^2. These are character traces for finite-order geometric symmetries. The identity, the three involutions and the two classes of four elements of order three give

$$\dim eU = \frac{8 + 3\cdot 0 + 4\cdot 2 + 4\cdot 2}{12} = 2.$$

Write $W = eU$ and $K = (e_C - e)U$. The latter also has dimension two. The line of type $(3,0)$ in U is fixed pointwise by d by [18, §5.1]. The entire A_4 preserves this Hodge line, so its action there factors through $A_4^{\mathrm{ab}} \simeq C_3$, generated by the class of d. Thus W has types $(3,0), (0,3)$, and K has types $(2,1), (1,2)$. The group acts by isometries of the natural alternating pairing (cup product on parabolic cohomology). The averages are self-adjoint, so the restrictions to W and K are nondegenerate and both determinants are $\mathbf{Q}_\ell(-3)$.

B Frobenius and the torus count

For $p \geq 5$ the reduction is good. In Hesse coordinates the two sets of four singular fibers remain disjoint: the only possible finite overlap between μ_3 and $-2\mu_3$ would imply $1 = -8$. At these primes the projectors (4) have p-integral coefficients. Their rational crystalline realizations are weakly admissible and compatible with the good-prime traces on the ℓ-adic realization. In particular

$$v_p(\mathrm{Tr}(\mathrm{Frob}_p \mid K)) \geq 1. \tag{5}$$

Indeed K has Hodge slopes 1 and 2. The Newton polygon lies above the Hodge polygon, so each Frobenius eigenvalue has valuation at least one. Their product equals p^3. This observation also distinguishes the factors whenever only one of the two candidate traces is divisible by p.

B.1 Trace on the four-dimensional constituent

On the open Hesse pencil define

$$(x, y, z) = \frac{(X^3, Y^3, Z^3)}{3sXYZ}.$$

Then $x + y + z = 1$, $xyz = u = 1/(27s^3)$. On the paired pencil the corresponding parameter is $v = c/u$. This is a degree-three isogeny after a choice of origin (equivalently, a finite map between the smooth genus-one compactifications inducing an isomorphism on rational first cohomology). Indeed the three preimages form an orbit of the order-three diagonal projectivity $(X, Y, Z) \mapsto (X, \zeta Y, \zeta^2 Z)$ on the generic fiber. The target coordinates are unchanged by $s \mapsto \zeta s$ with its specified lift D_0, and the paired coordinates by its conjugate. Taking the invariants of this cyclic base cover consequently identifies V with the parabolic cohomology in the u coordinate of the product of the two elliptic local systems. One may check the descent on the open locus and then extend by invariant stalks. Finite pushforward and projection onto invariants commute with this extension in characteristic zero.

Write $N(u)$ for the number of triples in $(\mathbf{F}_p^\times)^3$ whose sum is 1 and product is u. The cubic compactification $xy(w - x - y) = uw^3$ adds three rational points at infinity. Its smooth trace and its nodal invariant-stalk trace at $u = 1/27$ therefore both equal $t(u) = p - 2 - N(u)$ for $u \in \mathbf{F}_p^\times$. For the nodal cubic this follows either by normalization or by counting its smooth projective model with the two branches identified. There is no invariant stalk of the product at 0 or ∞. To see this, a Weierstrass model for the first elliptic system is $Y^2 + XY + uY = X^3$, obtained with $X = -u/x$ and $Y = -Xy$. Its invariants are $c_4 = 1 - 24u$ and $\Delta = u^3(1 - 27u)$. Thus at $u = 0$ its monodromy is unipotent. At infinity the substitutions $h = 1/u$, $X = h^{-2}X'$, $Y = h^{-3}Y'$ give minimal discriminant of order eight and c_4 of order three (Kodaira type IV*). Its tame inertia on first cohomology has eigenvalues ζ, ζ^2. In the paired system these roles are interchanged at the two points. At the two distinct nonzero nodes only one factor is singular. The trace formula, with no global invariants or dual invariants, gives

$$\begin{aligned} T_p := \operatorname{Tr}(\mathrm{Frob}_p \mid V) &= -\sum_{u \in \mathbf{F}_p^\times} t(u)t(c/u) \\ &= p^3 - 5p^2 + 10p - 8 - M_p, \qquad (6) \\ M_p &= \sum_{u \in \mathbf{F}_p^\times} N(u)N(c/u). \end{aligned}$$

In obtaining (6) we have used $\sum_u N(u) = p^2 - 3p + 3$, by inclusion and exclusion for the three affine lines $x = 0$, $y = 0$ and $x + y = 1$.

B.2 Extracting the truncated period

Here M_p counts ordered pairs of triples. On six nonzero coordinates with the prescribed two sums, write u_0 for their total product. Modulo p the indicator of $u_0 = c$ is $-\sum_{j=1}^{p-1}(u_0/c)^j$. Put $B_j = \sum_{x+y+z=1,\, xyz \neq 0}(xyz)^j$. Then

$$M_p \equiv -\sum_{j=1}^{p-1} c^{-j} B_j^2 \pmod{p}.$$

For $j = p-1$ we have $B_j \equiv 3$. For $1 \le j \le p-2$ put $k = p-1-j$. Expanding $x^j y^j (1-x-y)^j$ and summing over all x, y leaves only the monomial with added exponents k, k. It occurs exactly when $3k \le p-1$, and has coefficient

$$\frac{(p-1-k)!}{(k!)^2(p-1-3k)!} \equiv A_k \pmod{p}.$$

The two nonzero power sums each contribute -1. For the remaining k the coefficient A_k is zero modulo p. It follows that

$$M_p \equiv -8 - S_p \pmod{p}, \qquad T_p \equiv S_p \pmod{p}. \tag{7}$$

In particular, with the identification of W with f, equations (5)–(7) give $S_p \equiv a_p \pmod{p}$ at every good prime, including any nonordinary prime.

C Modular identification

Let g denote the normalized newform 54.2.a.a with good-prime coefficients b_p.

Theorem C.1. *For the rank-four constituent of the specified Hesse model, the semisimplifications satisfy*

$$V_{c,\ell}^{\mathrm{ss}} \simeq \rho_{f,\ell} \oplus \rho_{g,\ell}(-1).$$

The good-prime Euler polynomial of V is therefore $(1 - a_p T + p^3 T^2)(1 - p b_p T + p^3 T^2)$.

Proof. Compare W_2 with $\rho_{f,2}$ and $K_2(1)$ with $\rho_{g,2}$. Each pair has the same determinant by the alternating pairing and the trivial nebentypus of the newforms. Their ramification is confined to $2, 3$: the two sets of bad Hesse fibers are disjoint and tame over $\mathbf{Z}[1/6]$ after adjoining ζ, the rational elliptic surfaces are smooth there, and the fiber product has good reduction. Compactness gives stable $\mathbf{Z}_2$-lattices even though the averaging operators were defined over $\mathbf{Q}_2$.

The four polynomials can be determined by point counts over $\mathbf{F}_p$ and $\mathbf{F}_{p^2}$. Write $t_q = \mathrm{Tr}(\mathrm{Frob}_q \mid V)$. The trace-count identity (6) also applies over these extensions. The reciprocal quartic has coefficients $1, -t_p, (t_p^2 - t_{p^2})/2, -p^3 t_p, p^6$ in ascending degrees. The following exact counts are reproducible by the accompanying `verify_four_primes.py`. They agree with Bönisch's computations in [18, §5.1].

p	t_p	t_{p^2}	$\mathrm{Tr}\, W$	$\mathrm{Tr}\, K$	a_p	b_p
5	18	−266	3	15	3	3
7	22	−482	29	−7	29	−1
13	−32	−5684	20	−52	20	−4
19	−68	−14756	−106	38	−106	2

Here is why the indicated ordering of factors is forced. The two displayed reciprocal pairs have distinct traces and each has product p^3. A root of the quartic has just one reciprocal partner, so its two rank-two summands have to be those pairs. The minimum Hodge slope of K is one. Crystalline comparison at the four good primes and the Newton–Hodge inequality therefore give $v_p(\lambda) \ge 1$ for both of its roots. Precisely one of the two candidate traces is divisible by p. The use of the same projected factor across realizations follows from the geometric action: the two projectors are rational combinations of automorphisms on the parabolic term. More explicitly, on the smooth projective fiber product one takes the alternating trace of an automorphism composed with Frobenius by intersection with the diagonal. Degrees $0, 2, 4, 6$ are algebraic and degrees $1, 5$ vanish. To obtain

the primitive term in degree three one removes the ruled-fiber contribution, given by the reduced component spaces tensored with the elliptic $H^1(-1)$. This calculation is common to the crystalline and ℓ-adic realizations. If Frobenius permutes automorphisms or components, take the trace of the entire averaged correspondence. The last two columns are the Fourier coefficients of the labeled newforms [18, §5.1].

We specify the finite comparison criterion, since agreement of traces at an unspecified set of primes would not suffice. For each of the four representations, the residual characteristic polynomials at $5, 7, 13$ are X^2+X+1, X^2+X+1, X^2+1 respectively. Li, Long and Yang, Lemma 6.1 in [19], show that these data and the ramification bound determine a unique surjective residual representation up to conjugacy. Its field is the splitting field of $X^3 + 3X - 2$, with quadratic subfield $\mathbf{Q}(\sqrt{-6})$. Their Theorem 6.1 and its field analysis in the proof of Theorem 6.2 now apply to each pair. In detail, a nonzero deviation between equal-determinant lifts would be detected at a Frobenius of order at least four in one of the deviation groups $S_4, S_4 \times C_2, S_3 \times C_2$. For the first two possibilities the three admissible quartic fields have defining polynomials

$$X^4 - 4X - 3, \qquad X^4 - 8X + 6, \qquad X^4 - 12X^2 - 16X + 12.$$

The first two yield a four-cycle at 13, the third at 19. In the remaining case an independent quadratic character unramified outside $2, 3$ is nontrivial at 5 or 7: among the squareclasses generated by $-1, 2, 3$ only $1, -6$ split at both primes. The residual Frobenius there has order three, giving an element of order six. The table has equal traces for each comparison at all four primes, eliminating these deviations. One may extend scalars to $\mathbf{Q}_2(i)$ to use precisely the coefficient ring of [19]; its residue field is still $\mathbf{F}_2$.

This proves both comparisons at $\ell = 2$, hence the asserted Euler polynomials at every $p \geq 5$. The common Frobenius polynomials for V across ℓ, followed by Chebotarev and Brauer–Nesbitt, give the stated semisimplifications for all ℓ. □

D Finite product identities for the two higher digits

By Theorem C.1 and (7) the first digit is established for all good primes. The congruence modulo p^3 is equivalent, without an ordinarity hypothesis, to the scalar condition

$$\frac{M_p + 8 + S_p}{p} + b_p - 10 + 5p \equiv 0 \pmod{p^2}. \tag{8}$$

The left quotient is integral by (7). The equivalence follows by substituting $a_p = T_p - pb_p$ in (6).

D.1 Jacobi sums and factorials

Let $p \geq 5$, $c = -1/5832$, and $T_p = \mathrm{Tr}(\mathrm{Frob}_p \mid V_{z_*}) = a_p + pb_p$. Set $m = \lfloor (p-1)/3 \rfloor$, $n = \lfloor 2(p-1)/3 \rfloor$, and, for $0 \leq k \leq m$, put

$$\delta_k = \frac{pk}{p-1}, \qquad W_k = \frac{\prod_{j=1}^{3k}(j - 3\delta_k)}{\prod_{j=1}^{k}(j - \delta_k)^3}.$$

All the denominators in W_k are p-adic units. For $m < k \leq n$ set $r_k = 3k - (p-1)$. Define the following explicitly computable p-adic integer:

$$B_p = \frac{1}{1-p}\left(\sum_{k=0}^{m} c^{p^2 k} W_k^2 + p^2 \sum_{k=m+1}^{n} c^k \left(\frac{r_k!}{(k!)^3}\right)^2\right).$$

Proposition D.1. *The good-prime trace satisfies $T_p \equiv B_p \pmod{p^3}$. In particular, $B_p - S_p \in p\mathbf{Z}_p$, and the congruence $S_p \equiv a_p \pmod{p^3}$ for a given p is equivalent to*

$$\frac{B_p - S_p}{p} \equiv b_p \pmod{p^2}.$$

Write $H_j = \sum_{i=1}^{j} i^{-1}$, $H_0 = 0$, and $q_p(c) = (c^{p-1} - 1)/p$. At precision p^2 one obtains the equivalent identity in $\mathbf{F}_p$

$$b_p = \sum_{k=0}^{m} A_k^2 c^k \left[1 + k q_p(c) + 6k(H_{3k} - H_k)\right].$$

Proof. For $u \in \mathbf{F}_p^\times$ put

$$N_u = \#\{(x, y, w) \in (\mathbf{F}_p^\times)^3 : x + y + w = 1,\ xyw = u\}, \quad h(u) = p - 2 - N_u.$$

The toric trace formula gives $T_p = -\sum_{u \neq 0} h(u) h(c/u)$. We have $\sum_{u \neq 0} N_u = p^2 - 3p + 3$, so $\sum_{u \neq 0} h(u) = -1$. For a nontrivial multiplicative character χ, its Mellin coefficient $\sum_u h(u)\chi(u)$ is $-J(\chi, \chi, \chi)$, by the defining equation for N_u. With ω the Teichmüller character, write $J_k = J(\omega^{-k}, \omega^{-k}, \omega^{-k})$ for $1 \le k \le p - 2$. Character orthogonality gives the exact identity

$$T_p = \frac{1}{1 - p}\left(1 + \sum_{k=1}^{p-2} \omega(c)^k J_k^2\right).$$

For each $k \ge 1$ write $3k = e(p-1) + r$ with $1 \le r \le p-1$, and put $x = k/(p-1)$. The Gross–Koblitz formula (B. H. Gross and N. Koblitz, *Gauss sums and the p-adic Γ-function*, Ann. of Math. 109 (1979), 569–581, Theorem 1.7) for the usual Gauss sum is $g(\omega^{-k}) = -\pi^k \Gamma_p(x)$ with $\pi^{p-1} = -p$. The Gauss–Jacobi sum relation gives

$$J_k^2 = p^{2e}\left(\frac{\Gamma_p(x)^3}{\Gamma_p(r/(p-1))}\right)^2.$$

Indeed, for $r < p - 1$ the unsquared Jacobi sum is $g(\omega^{-k})^3/g(\omega^{-r})$. For $r = p - 1$ it is $-g(\omega^{-k})^3/p$, which gives the same squared expression since $\Gamma_p(1) = -1$. This convention includes both integral break points of $3k/(p-1)$. All terms with $e \ge 2$ vanish modulo p^3.

For $e = 0$ write $x = -k + \delta_k$ and $3x = -3k + 3\delta_k$. Morita's functional equation and $3k \le p - 1$ give

$$\frac{\Gamma_p(x)^3}{\Gamma_p(3x)} = W_k \frac{\Gamma_p(\delta_k)^3}{\Gamma_p(3\delta_k)}.$$

For $v \in p\mathbf{Z}_p$ one has $\Gamma_p(v)^3/\Gamma_p(3v) \equiv 1 \pmod{p^3}$. To see this, each block $\prod_{j=1}^{p-1}(j + hp)$ is congruent to $(p-1)!$ modulo p^3 for integers $h \ge 0$: $H_{p-1} \equiv 0 \pmod{p^2}$ by Wolstenholme, and $\sum_{j=1}^{p-1} j^{-2} \equiv 0 \pmod{p}$. Thus $\Gamma_p(hp) \equiv \Gamma_p(p)^h \pmod{p^3}$ by Morita's product definition. The ratio at $v = hp$ equals 1 modulo p^3, and continuity gives the assertion for all $v \in p\mathbf{Z}_p$. For $e = 1$ the gamma ratio is needed only modulo p. Morita's function is constant modulo p on each residue class modulo p (this also follows by grouping its integral product into blocks). Since $x \equiv -k$ and $r/(p-1) \equiv -r \pmod{p}$, and since $\Gamma_p(-j) = 1/j!$ for $0 \le j \le p - 1$, this ratio is $r!/(k!)^3$ modulo p. Finally $\omega(c) \equiv c^{p^2} \pmod{p^3}$. These calculations give $T_p \equiv B_p \pmod{p^3}$.

For the second digit, $\delta_k \equiv -pk \pmod{p^2}$, and expansion of the finite products yields

$$W_k^2 \equiv A_k^2\left[1 + 6pk(H_{3k} - H_k)\right] \pmod{p^2}.$$

Also $c^{p^2k} \equiv c^k(1+pkq_p(c)) \pmod{p^2}$. The terms of S_p with $k > m$ are in $p^2\mathbf{Z}_p$. Expanding $(1-p)^{-1}$ now gives the displayed identity for the obstruction modulo p. The two equivalences follow by substituting $T_p = a_p + pb_p$. □

D.2 The elliptic trace as a second member of the same pencil

The curve with LMFDB label 54.a3 corresponds to g and has minimal equation $y^2 + xy = x^3 - x^2 + 12x + 8$. Its rational point $(1, 4)$ has order three [22]. The substitutions $x = 1 + X$, $y = 4 + X + Y$ give

$$Y^2 + 3XY + 9Y = X^3.$$

On replacing (X, Y) by $(9X', 27Y')$, one obtains $Y'^2 + X'Y' + \frac{1}{3}Y' = X'^3$. Thus the weight-two trace is precisely the trace $t(1/3)$ of the toric cubic in Appendix B. The coordinate changes have good reduction for $p \geq 5$.

In fact for each $u \in \mathbf{F}_p^\times$ Fourier inversion for a single copy of $N(u)$, with the coefficients B_j above, gives

$$N(u) \equiv -3 - \sum_{k=1}^{m} A_k u^k, \qquad t(u) \equiv \sum_{k=0}^{m} A_k u^k \pmod{p}.$$

Write $L_p = \sum_{k=0}^{m} A_k 3^{-k}$ in $\mathbf{F}_p$. The second-digit identity is therefore the purely finite congruence

$$\sum_{k=0}^{m} A_k^2 c^k \left[1 + kq_p(c) + 6k(H_{3k} - H_k)\right] = L_p \quad \text{in } \mathbf{F}_p. \tag{9}$$

This includes all good primes, with no hypothesis on a_p. For the third digit one can even recover the integer b_p from L_p: it is the unique integer b satisfying

$$b \equiv L_p \pmod{p}, \qquad b \equiv p + 1 \pmod{3}, \qquad |b| \leq 2\sqrt{p}.$$

Indeed the rational point of order three injects under good reduction, giving the second residue condition. The Hasse bound gives the interval, and $3p > 4\sqrt{p}$ ensures uniqueness. Consequently Proposition D.1 gives an all-prime criterion using only factorials and elementary arithmetic, independent of any tables of newforms. There is also a version which involves no reconstruction of an integer. Put $v = 1/3$ and $R_k = r_k!/(k!)^3$ on the middle range. By inversion for the single trace $h(v)$ the Mellin formula reads

$$b_p = \frac{1}{1-p}\left(1 + \sum_{k=1}^{p-2} \omega(v)^k J_k\right).$$

In the notation of the preceding proof the unsquared Jacobi sum equals $(-p)^e \Gamma_p(x)^3/\Gamma_p(r/(p-1))$, including $r = p - 1$. Reducing to modulus p^2 yields

$$(1-p)b_p \equiv \sum_{k=0}^{m} v^{pk} W_k - p \sum_{k=m+1}^{n} v^k R_k \pmod{p^2}.$$

Together with Proposition D.1, this proves that the good-prime conjecture is equivalent to the following single finite product identity for each $p \geq 5$:

$$\sum_{k=0}^{m} \left(c^{p^2k} W_k^2 - pv^{pk} W_k\right) + p^2 \sum_{k=m+1}^{n} \left(c^k R_k^2 + v^k R_k\right) \equiv (1-p)S_p \pmod{p^3}. \tag{10}$$

By Theorem 1.1 both (9) and (10) hold for every prime $p \geq 5$.

D.3 Computational details for the finite identities

The index of $\Gamma_0(54)$ is 108, with 12 cusps, genus four and no elliptic points. Thus $\dim S_4(\Gamma_0(54)) = 21$ and the Sturm bound is 36. In the basis of eta quotients used in Section 5, imposing $U_2 = 2$, $U_3 = 0$ and $T_5 = 3$ gives subspaces of dimensions $4, 3, 1$, respectively. At $p = 4999$ the normalized eigenform gives $a_p = -560608$ and the independent truncation gives $S_p \equiv 124924454391 \pmod{p^3}$. The Jacobi-sum and elliptic-trace calculations were checked separately by point counts at all 93 primes $5 \le p \le 500$.

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